PCR:

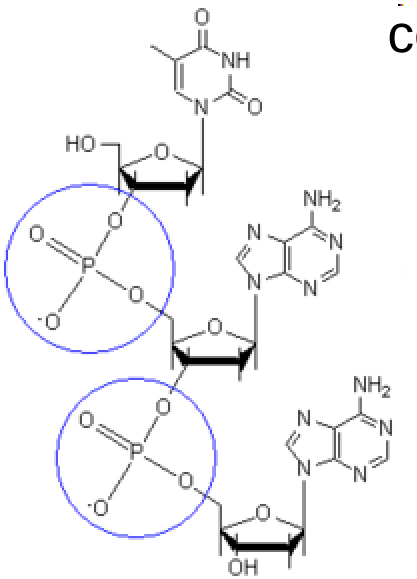
Replicate a sequence

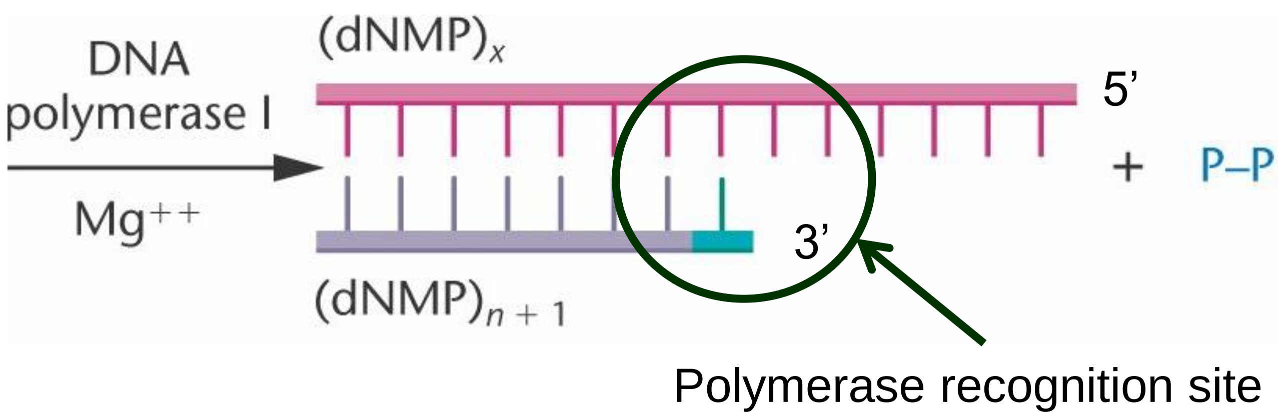
Copy + Paste

- Polymerase Chain Reaction
- Allows in-vitro amplification of DNA obtaining exact copies of the initial molecule.
- Invented by Kary Mullis in 1984 (nobel 1993), has been a main revolution in molecular biology.
- Uses precise temperature control to drive the reaction
- Uses:
  - Amplification of “signals” to enhance sensitivity of other methods
  - Selective amplification, to enrich targets.
  - Functional component for other applications

DNA polymerase

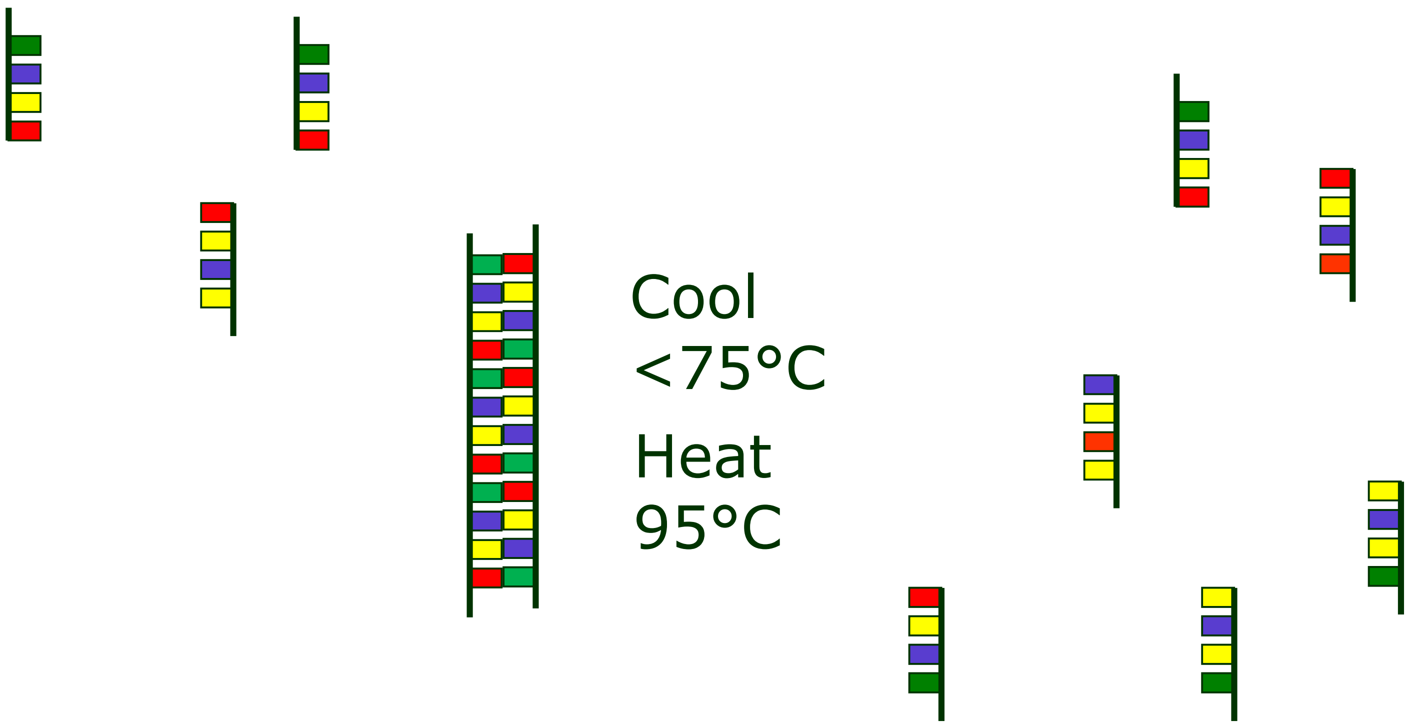
- In living cells has a key role in DNA duplication.
- Function:
  - Adds a nucleotide on 3’ termination of a DNA strand, when it is partially hybridized
  - The added nucleotide (unless error occurred) is the complementary of the corresponding one on the other strand.





Hybridization (revised)

- The same concept of Denaturation and Hybridization that allows microarray can happen in solution if the complementary DNA is not bonded to a surface



PCR in brief

- Emulates the 'in vivo' mechanism used by cells to replicate themselves.
- Allows the selection of the DNA portion to be amplified, defining the border sequences "brackets" (DNA primers).
- 3 main steps:
  1. Split the dsDNA into two ssDNA molecules
  2. Select the portion where the polymerase will operate by annealing the primers
  3. Polymerase actually replicates the missing strand

  Repeats the 3 steps (cycle)
- At each step the DNA doubles.

PCR Mix

- For the reaction to happen it is required in solution to be present:
  - DNA target to amplify, usually called ‘template’
  - The polymerase enzyme, in a thermal-stable version
  - The tri-phosphate bases (the building ‘bricks’)
  - A salt buffer with specific concentration
  - Magnesium ions, to stabilise pH levels
  - Couples of DNA Primers

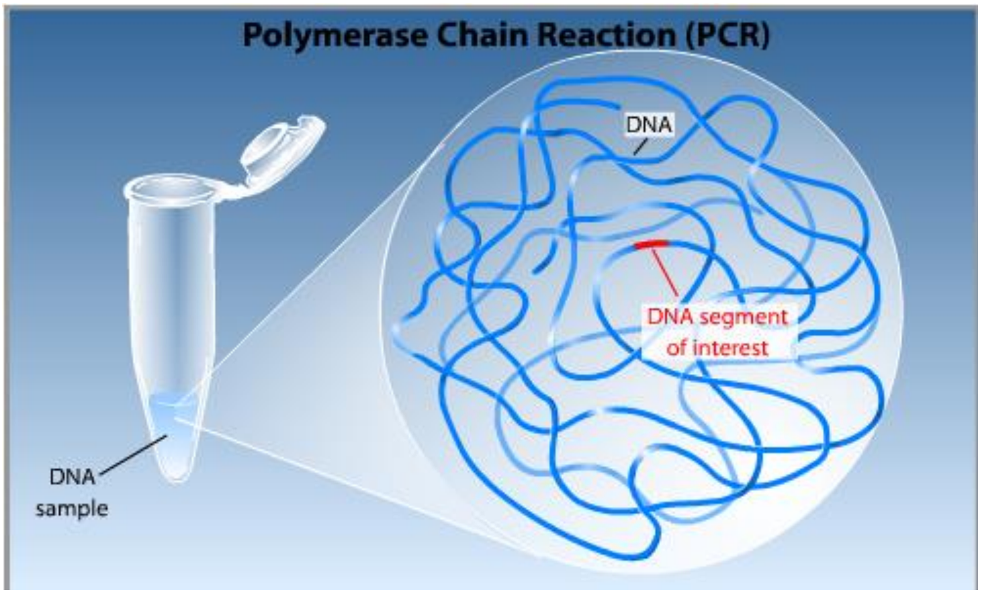
3 phases: Thermally driven:

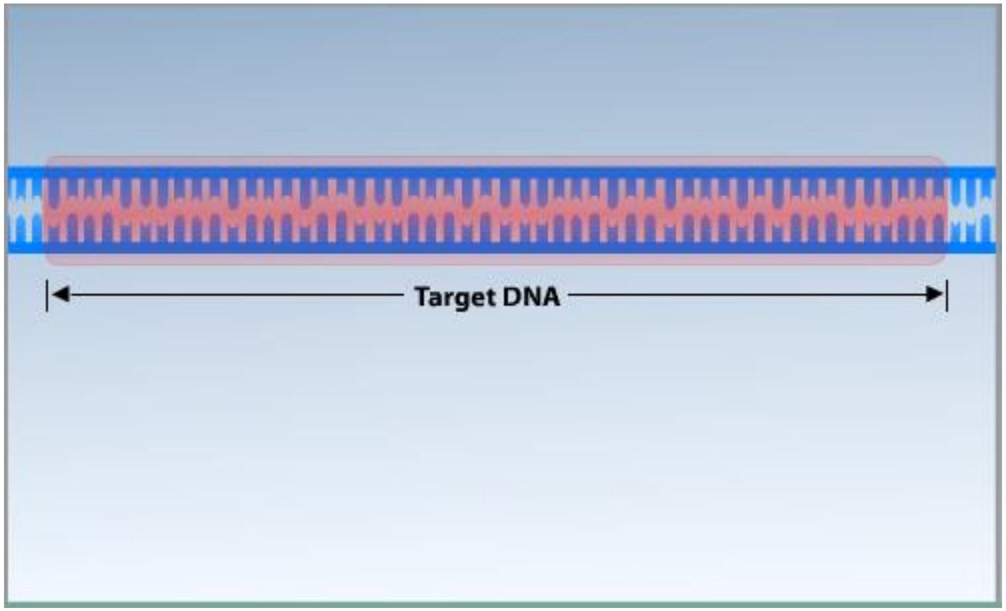
- Denaturation
  - At temperature around 100 C, the 2 DNA strands separate
- Annealing
  - At lower temperature (60 C), the single strands hibridize with primers, selecting the portion to be amplified
- Extension
  - At medium temperature (70 C), the polimerase start working, duplicating the DNA

The cycle repeats several times, every cycle the amount of DNA doubles

PCR Phases:

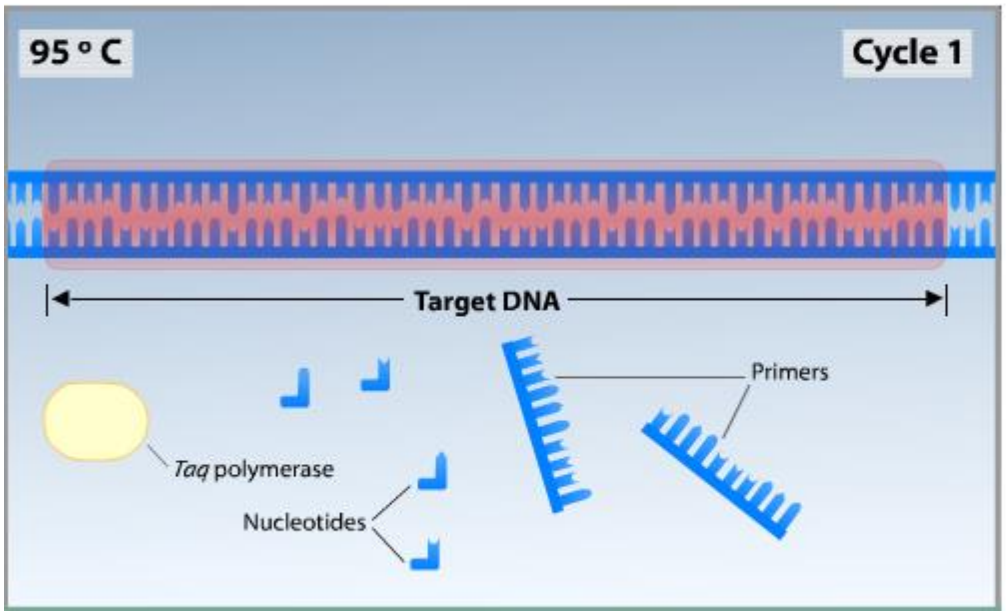
- Usually is not amplified the whole genome, but just a region of interest

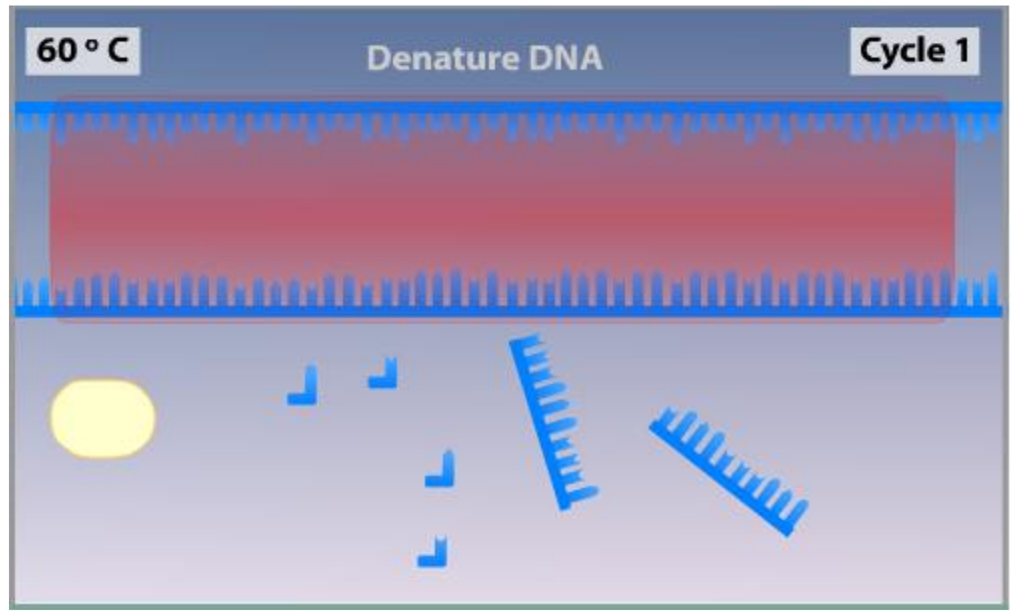




PCR Phases: /2

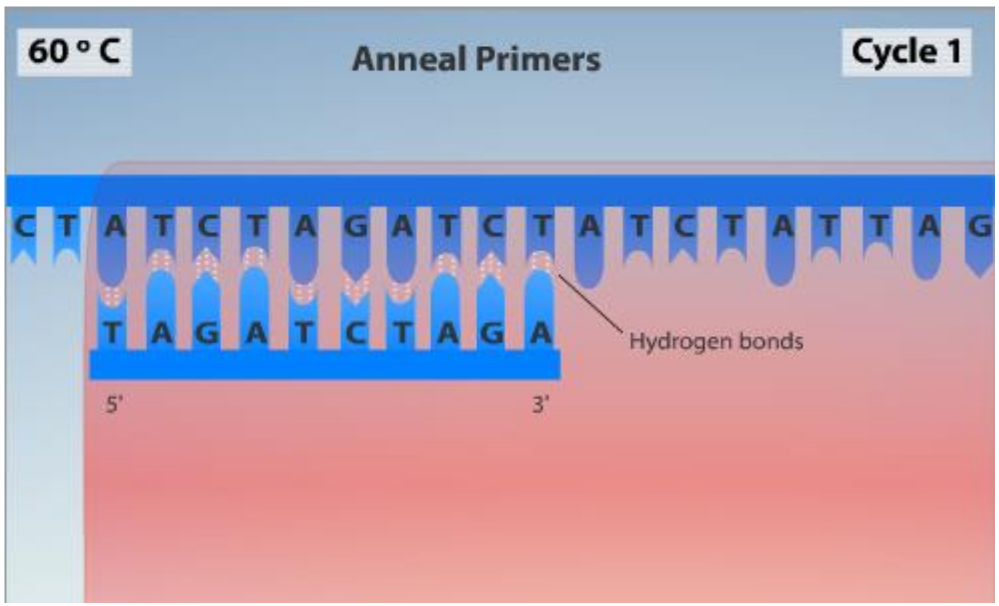
- Denaturation: Heating the system to 95 degrees, DNA separate into 2 strands





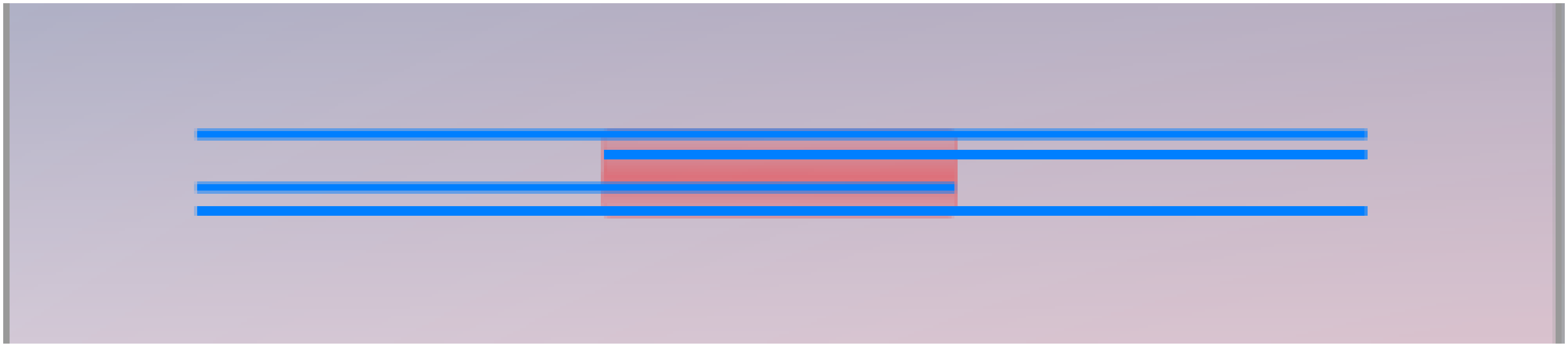
PCR Phases: /3

- Annealing: Cooling the system, primers find their complementary sequence and hybridise



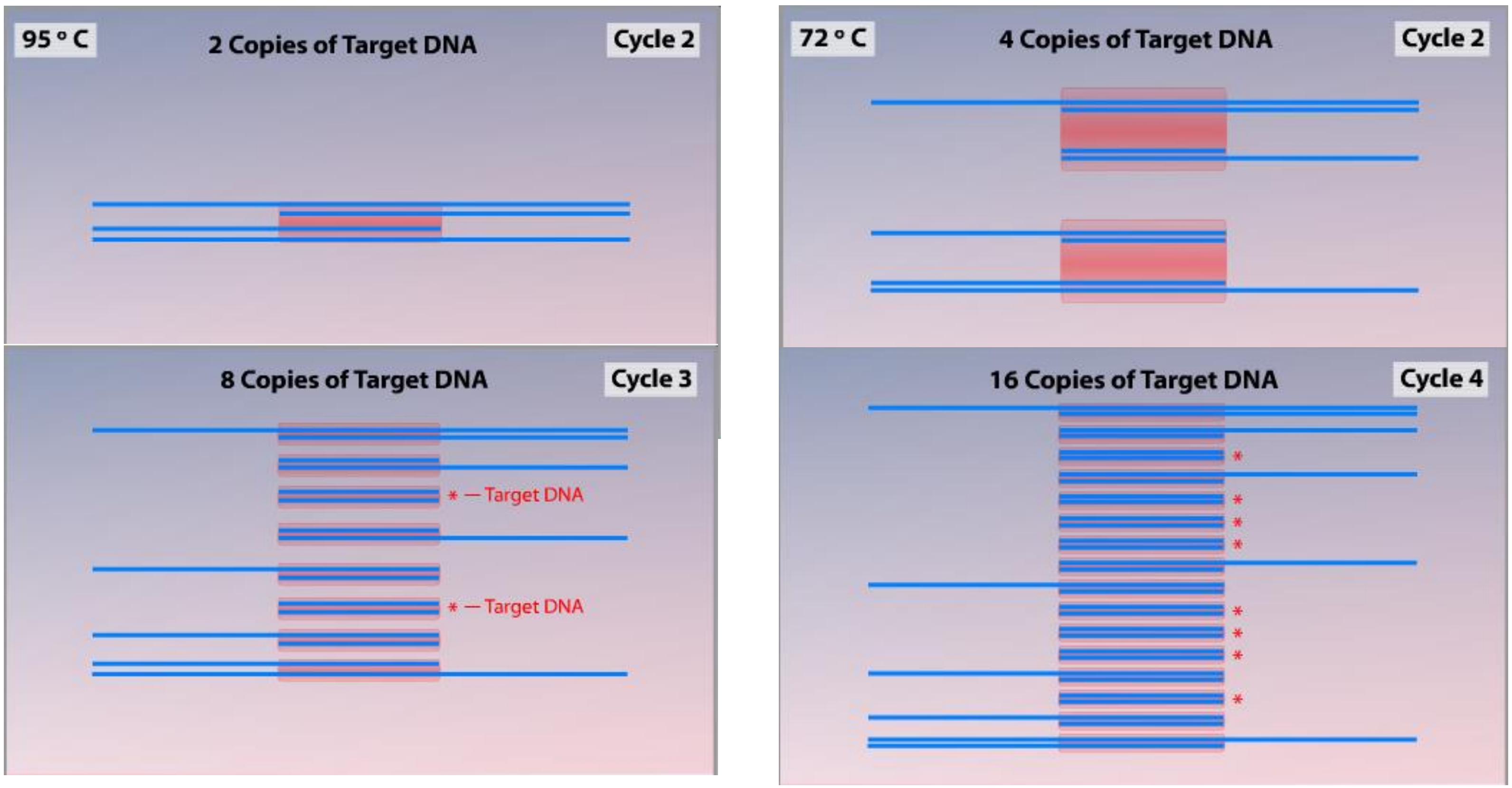
PCR Phases: /4

- Extension: the active phase of the reaction: starting from primers the new nucleotides are incorporated
- The temperature depends on the enzyme. Usually is about 72 degrees
- The result is a new DNA strand, starting from the primer and proceeding in 5’ to 3’ direction

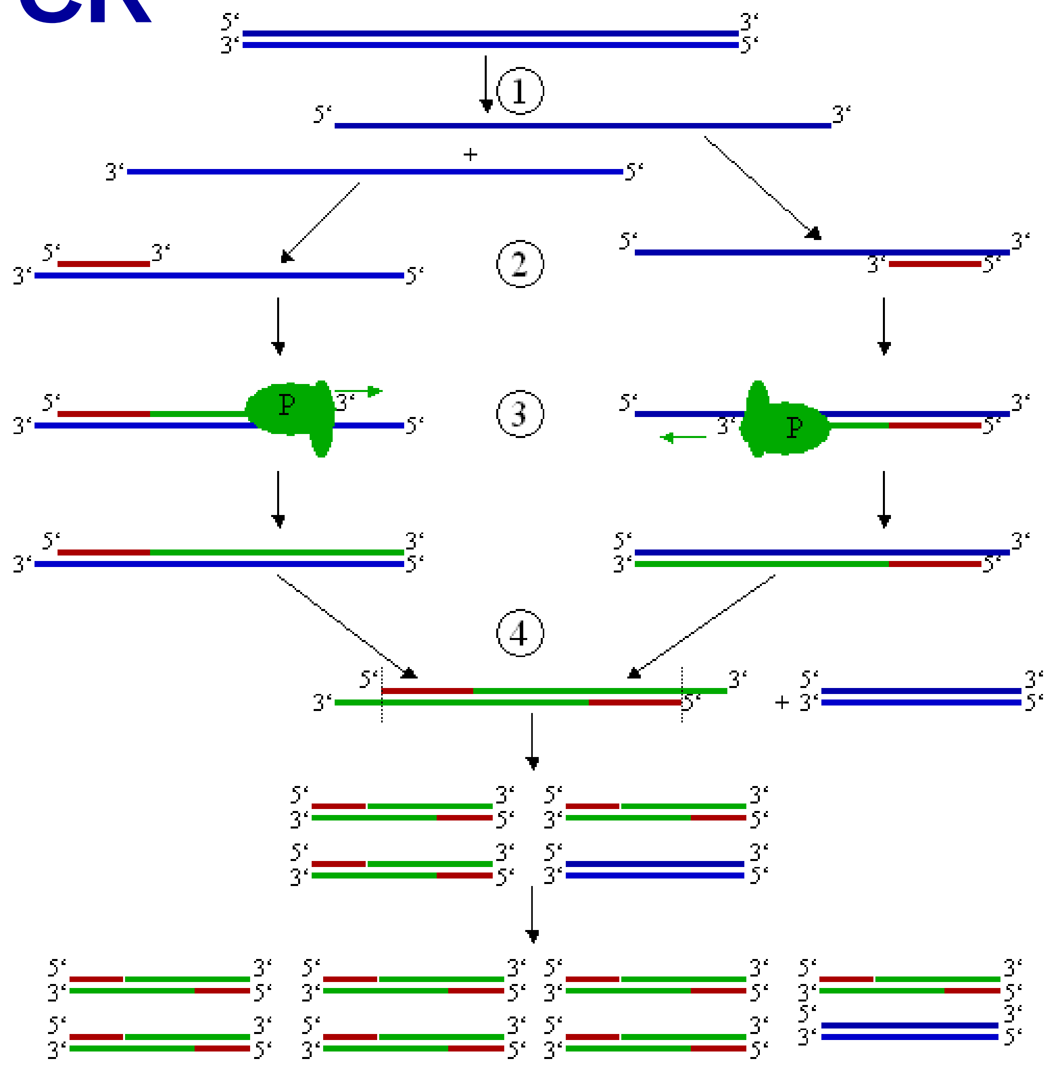


PCR Phases: /5

- Cycling: The steps are repeated sequentially, in each cycle DNA doubles.



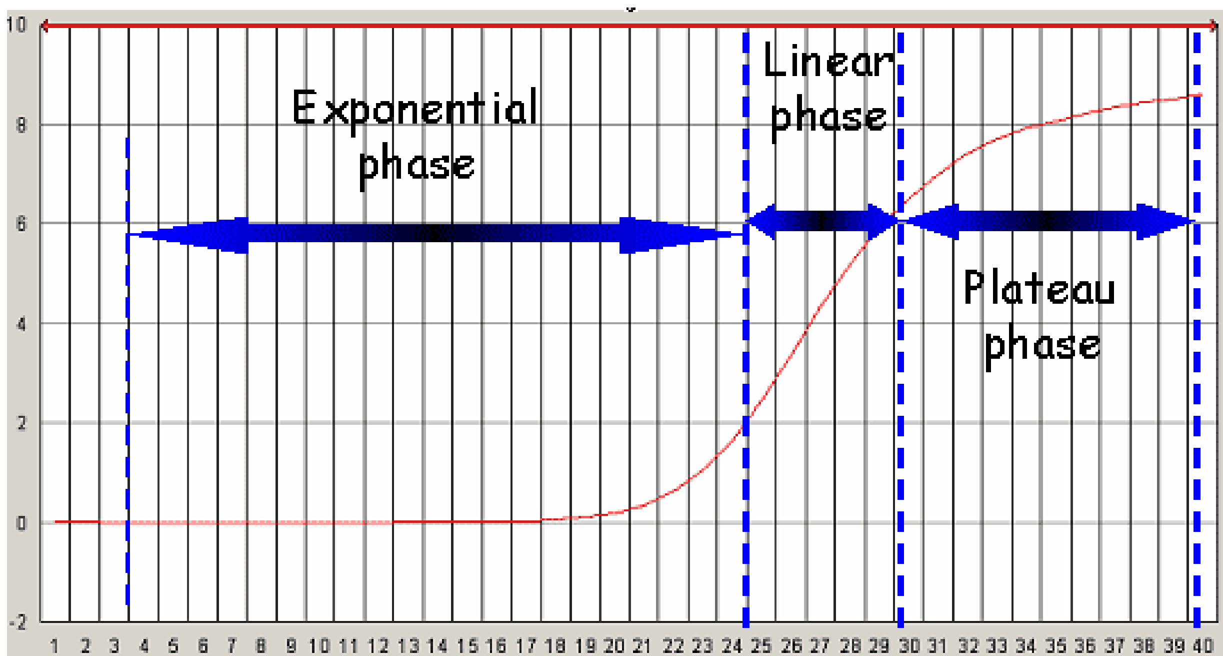
PCR



- After N cycles, for any initial DNA molecule we obtain:
  - 2 original strands
  - 2*N "half strand"
  - 2^N segments

PCR Dynamics

- During initial phase, the DNA quantity doubles every cycle, the reaction has an exponential behavior.
- When reagents start depleting, the growth slows down, becoming almost linear
- When the reagents exhaust, the reaction stops. (plateau)



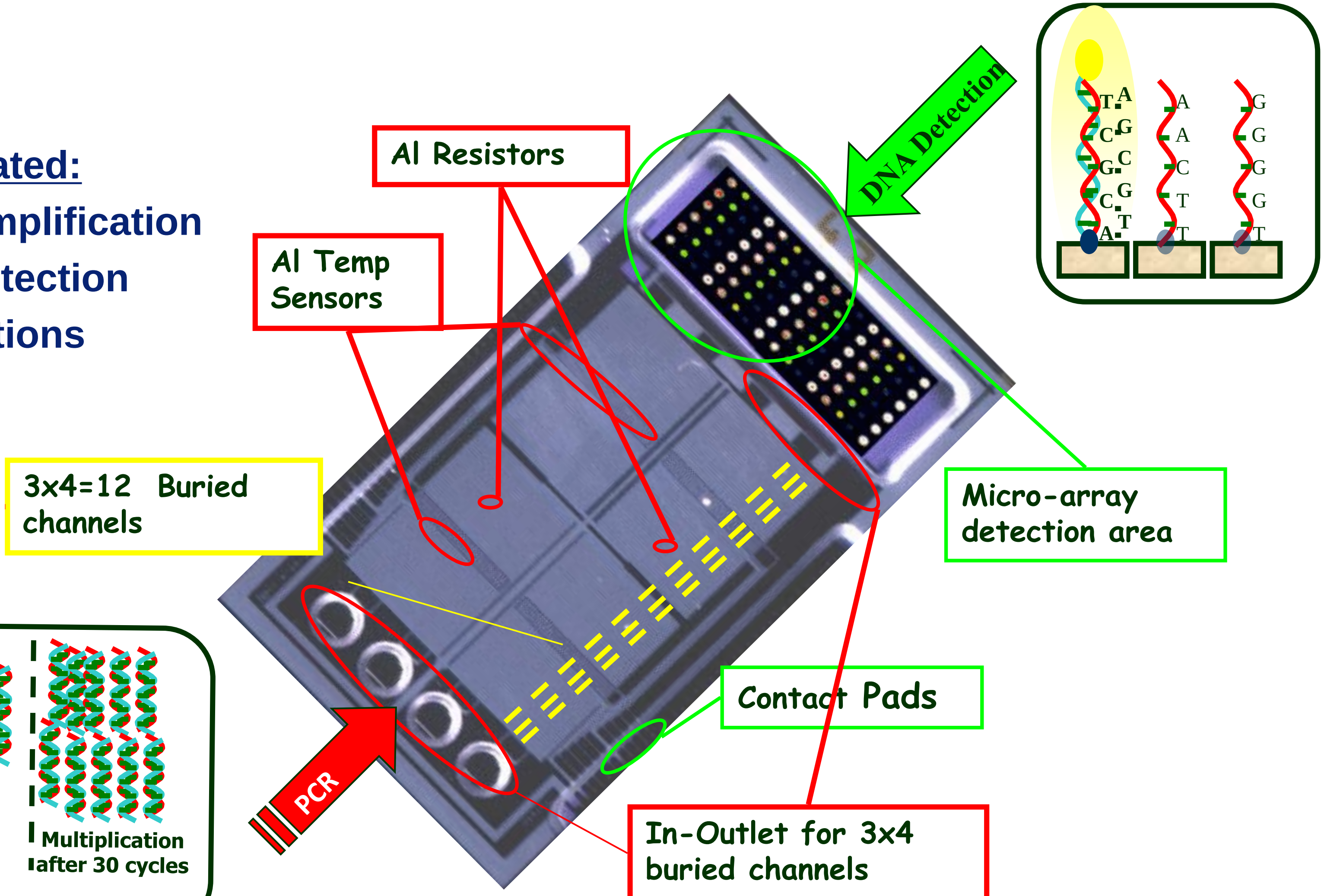
PCR to increase microarray sensitivity

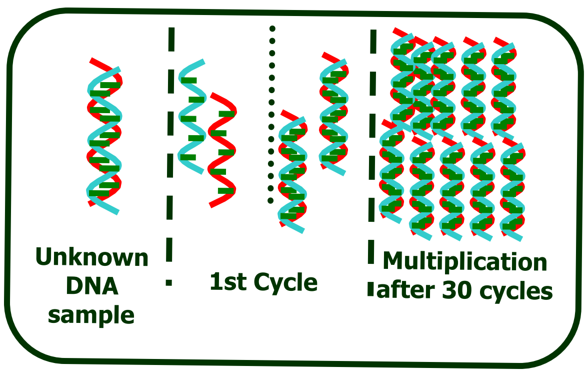
- Microarray signal is proportional to the DNA quantity
- If in the sample there is low DNA concentration, a direct hybridization may not be enough to detect the signal
- If a PCR is made before the microarray test, the potential DNA target is amplified and so become detectable.
- <u>Quantitative</u> information is completely <u>lost</u>

InCheck architecture

<u>Monolithically Integrated:</u>

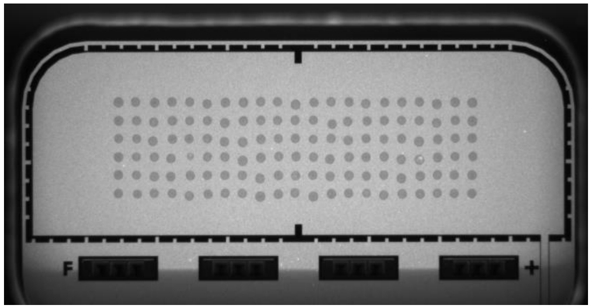
- **PCR based DNA Amplification**
- **Microarray DNA Detection**
- **Microfluidics Functions**



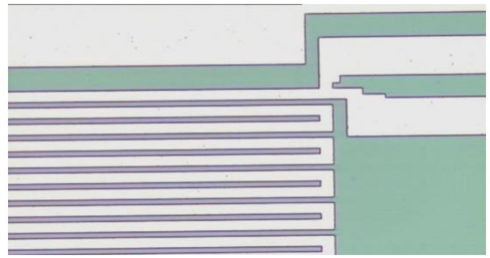


Lab On Chip functional elements

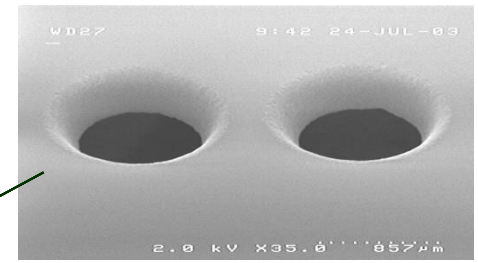
Microarray Detection Area

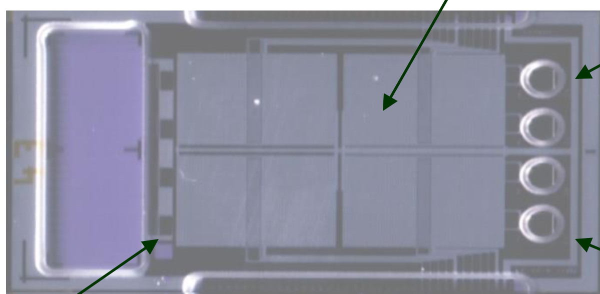


Thermal Sensors - Al/Si/Cu



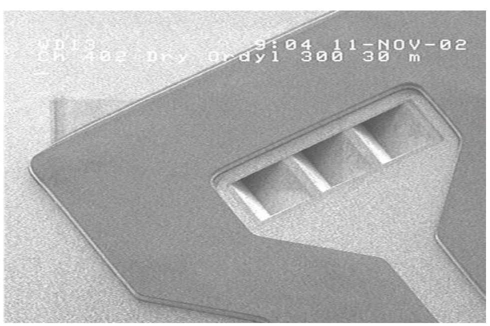
Fluidic Inlets - Glass



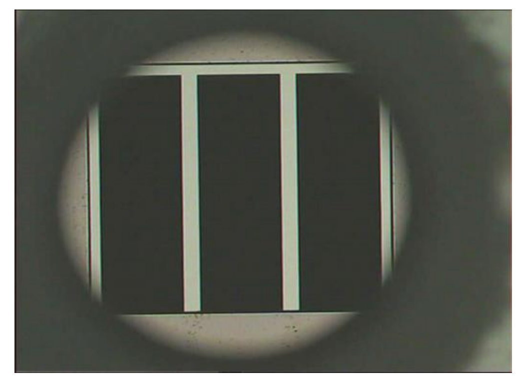


Core Silicon Chip

Surface Channels - Dry resist



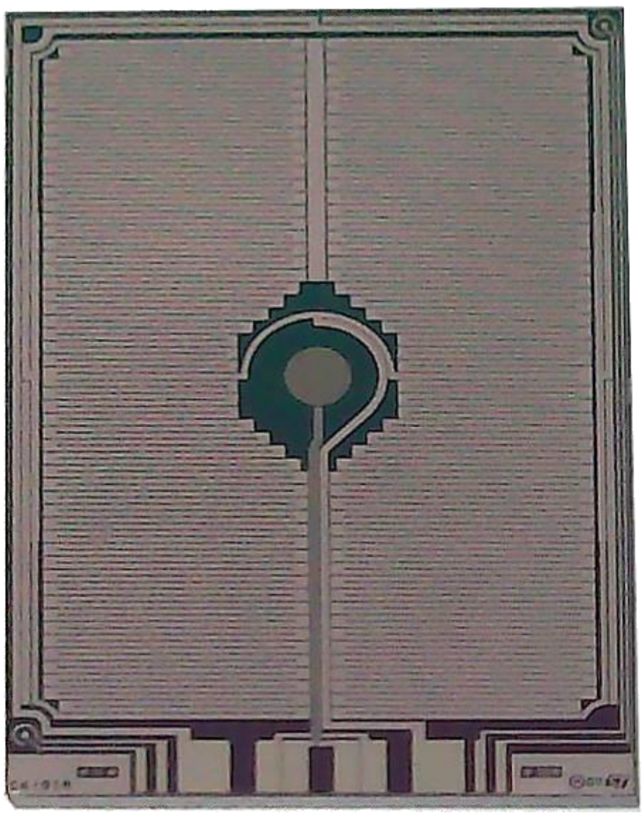
Inlet design – Buried Channels

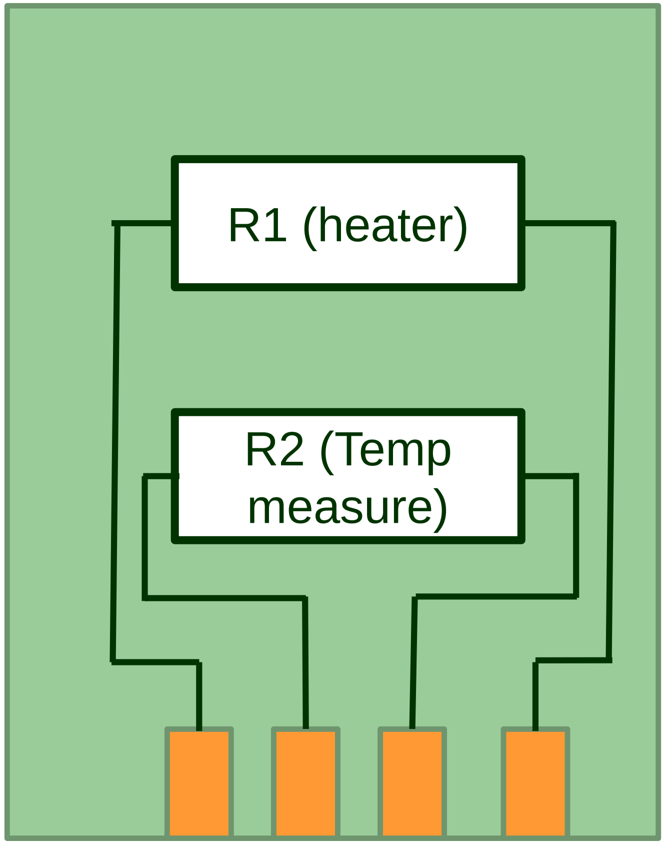


Importance of temperature precision

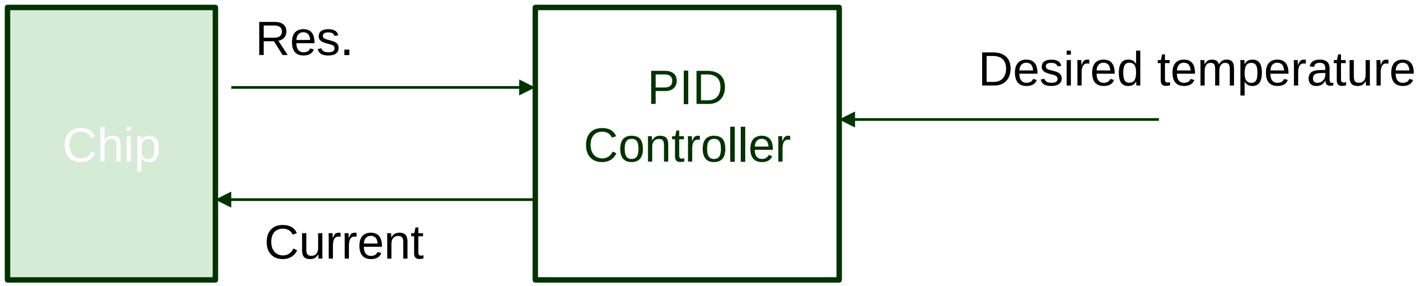
- T denaturation
  - Too high: polymerase can be damaged — No reaction
  - Too low: DNA does not denature completely — Low efficiency
- T hibridization
  - Too low: primers my bind a-specifically, amplifying unwanted DNA — Wrong result
  - Too high: Primers do not bind. — Low efficiency / No reaction
- T extension
  - Too high: primers may detach during extension — Low efficiency
  - Too low: reaction slows down. — Low efficiency / No reaction

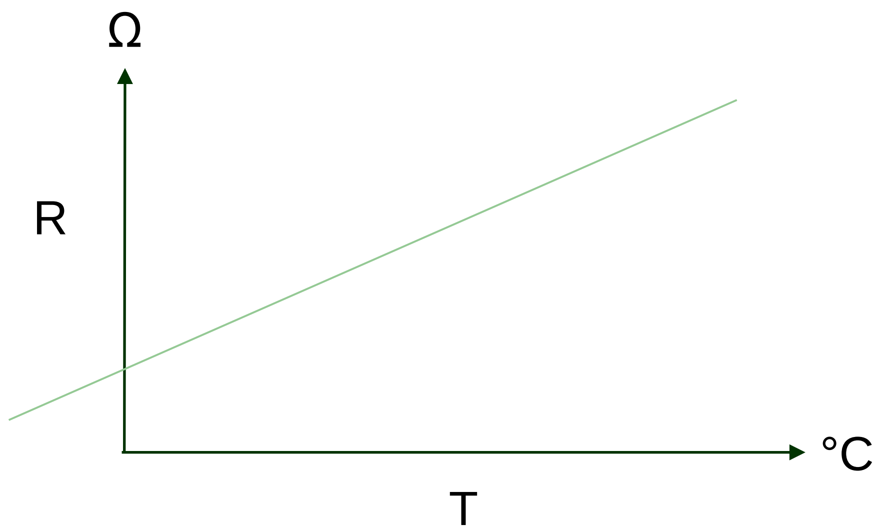
The chip is the thermal cycler





Closed loop control



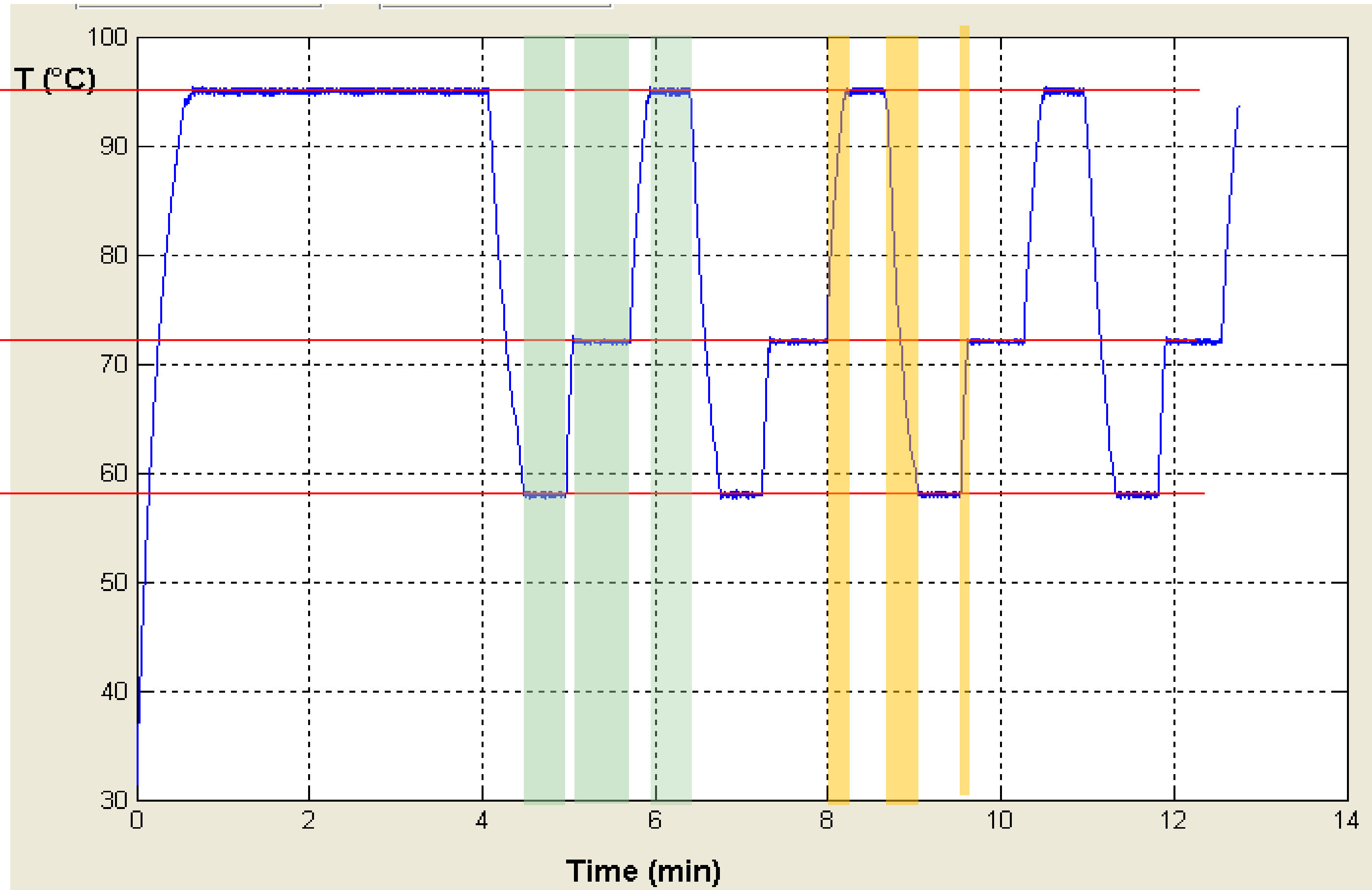


Temperature cycles

Denaturation

Extension

Annealing



Temperature Maintainment

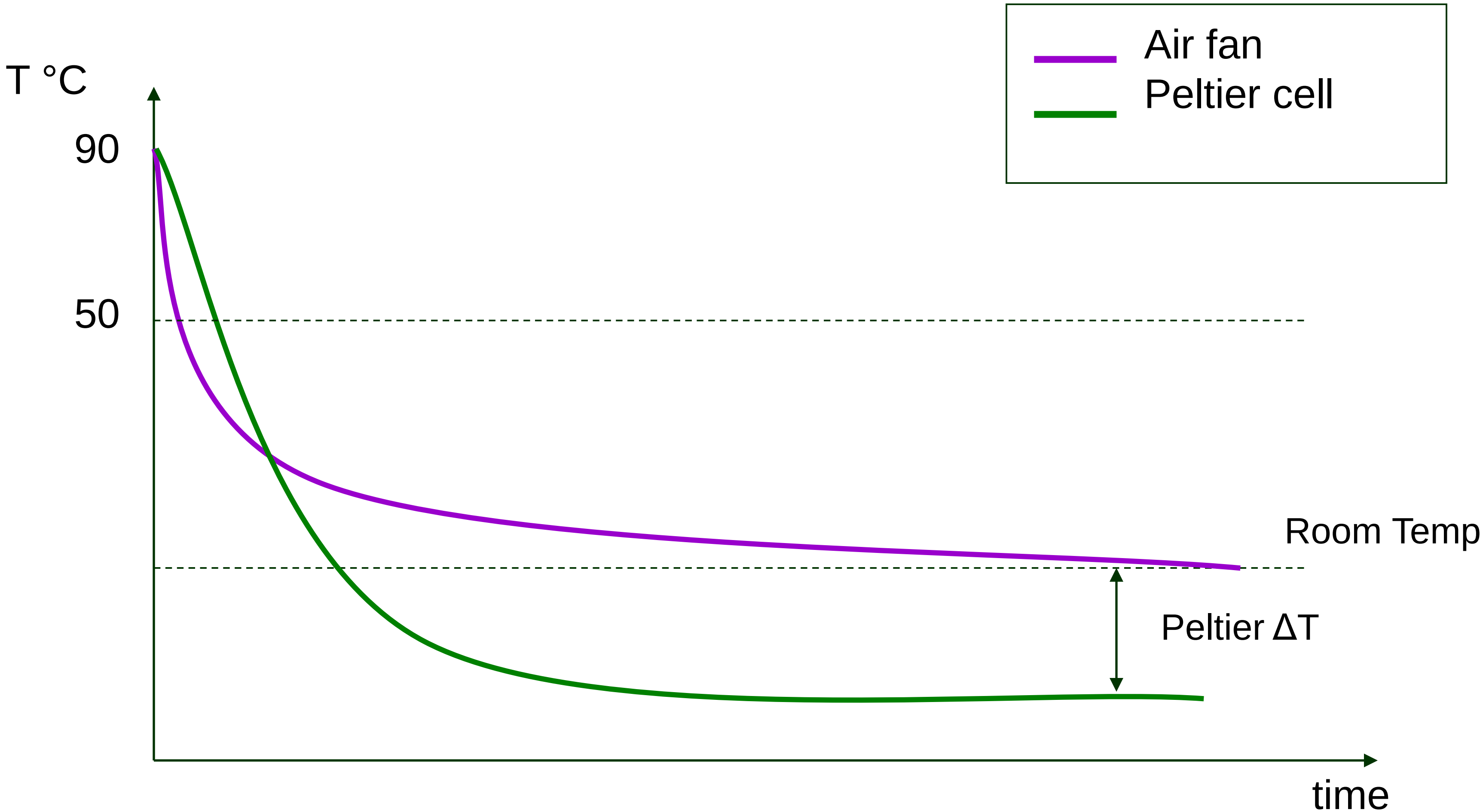
Temperature Change (ramps)

Importance of speed

- The maintaining times are defined by the PCR protocol: they depend on the reagents, the enzymes and the length of the target
- The ramps depend on the instrument: the faster the better.
- With standard (slow) enzymes, the total time spent for ramps is negligible, so there is not much gain in shortening it.
- With new enzymes, (fast) the ramp time is comparable with the maintaining time, so fast ramps shorten sensible the total test time.
- Combining fast enzymes and short ramps a typical test pass from 90 to 30 minutes. <u>For point of care it matters</u>

Cooling

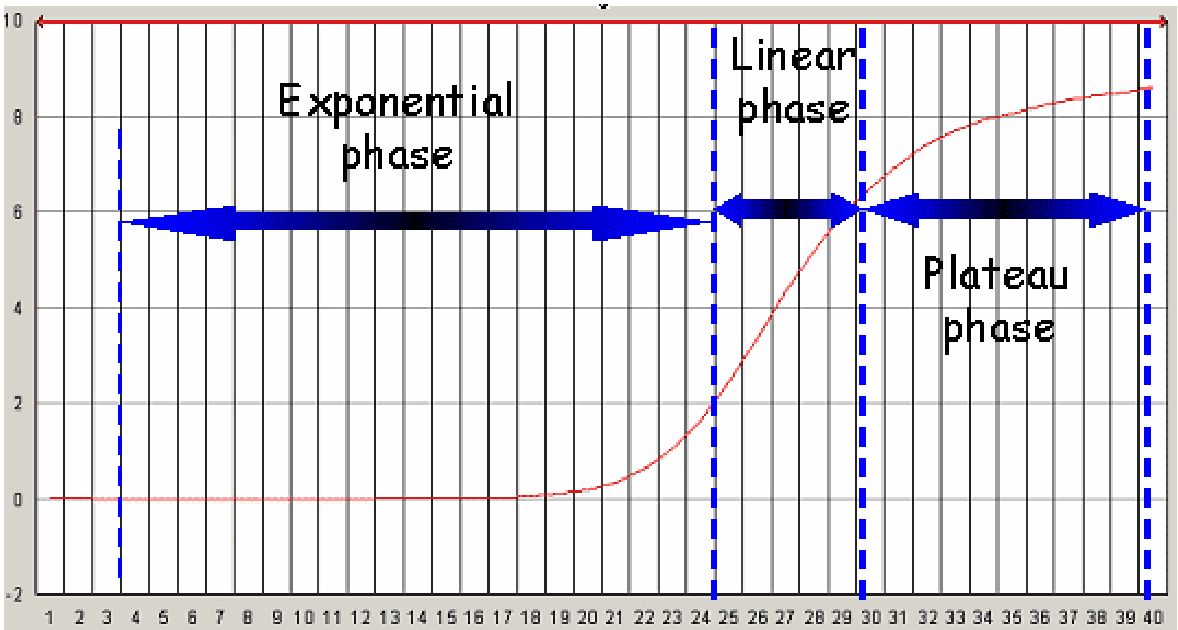
- The fastest the better!



Realtime PCR

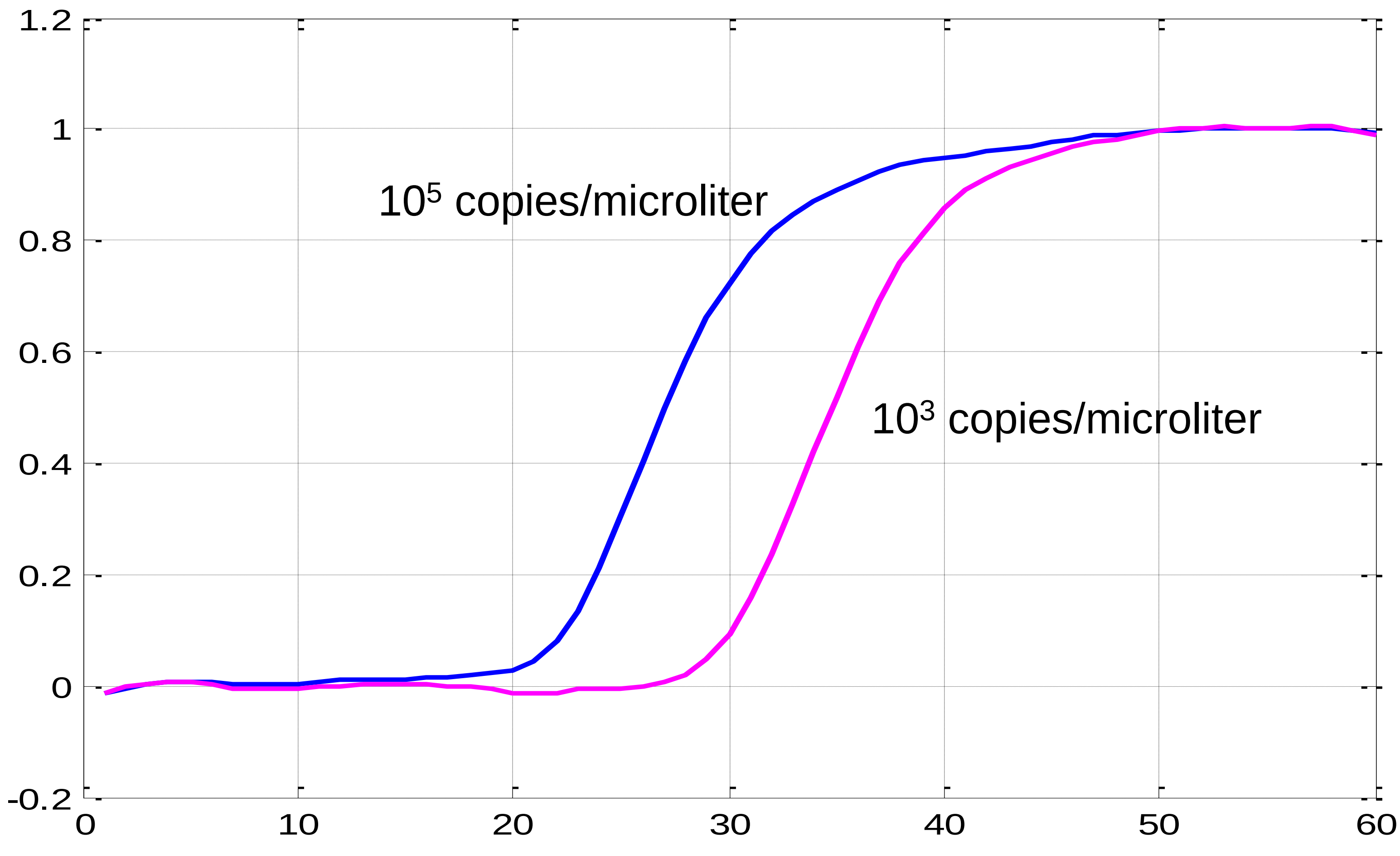
- Standard PCR allows the amplification in vitro of the DNA, to obtain identical copies of the initial molecule.
- The detection at endpoint (by microarray or electrophoresis) does <u>NOT</u> provide quantitative results.
- Realtime PCR allows a precise quantification of the DNA present in the initial sample.

PCR curve



- Plateau phase depends only upon the reagents
- The exponential phase depends upon the sample

DNA Concentration



Exponential phase

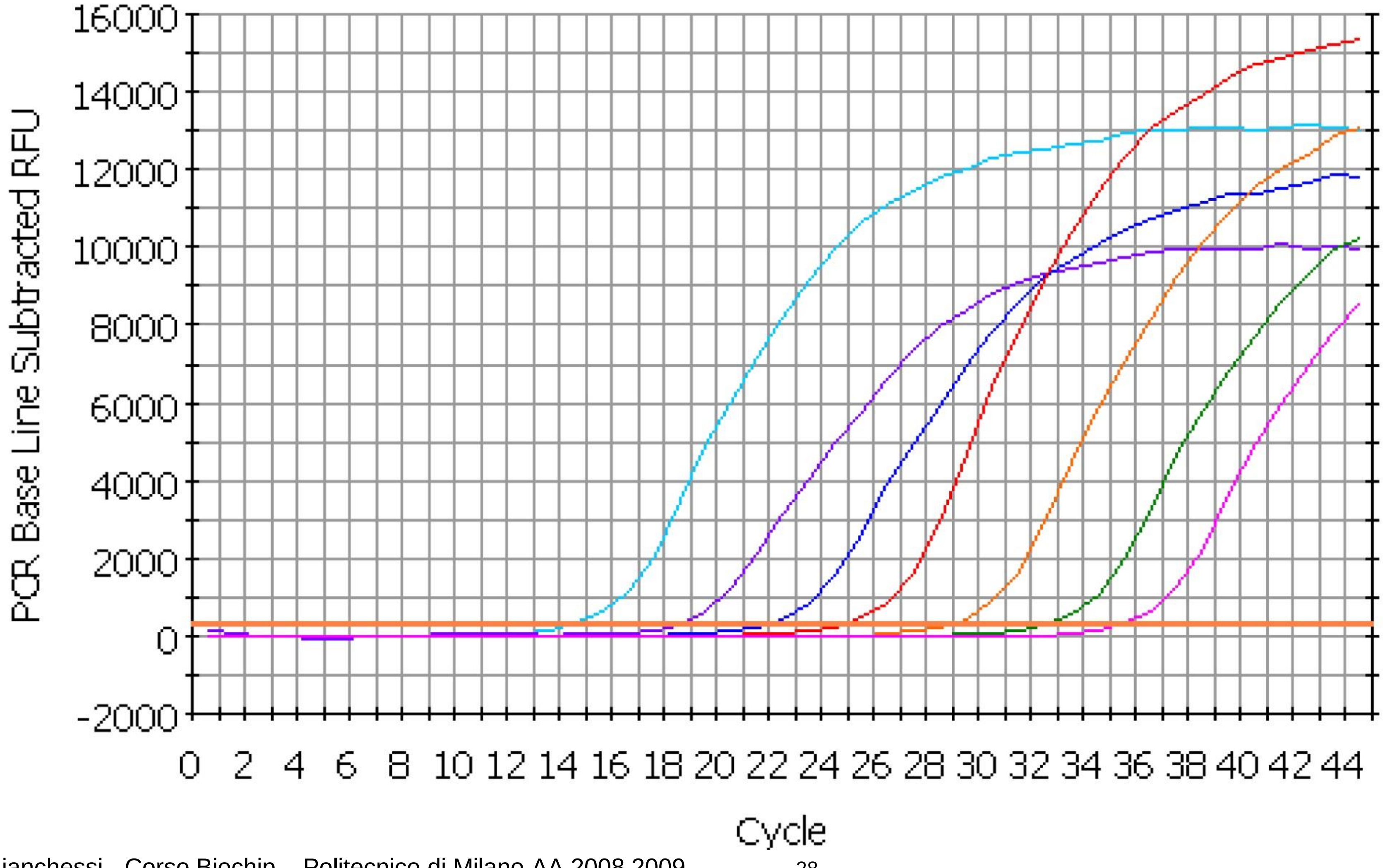
- $qDNA_{(N)} = Ci * 2^N$
  - $qDNA_{(N)}$ = Concentration of DNA at a certain cycle
  - Ci: Initial concentration
  - N=number of cycle

- $Ci * 2^N = 2^{N+\log 2\,(Ci)}$ (Just mathematics)
- So:
  - When Ci=1 : $qDNA_{(N)} = 2^N$
  - When Ci ≠1 : $qDNA_{(N)} = Ci * 2^M$ where M=N+log2 (Ci)
  - The initial concentration has the unique effect of horizontally translate the curve, that is always an exponential in base 2.
  - The translation factor is $-Log_2(Ci)$.

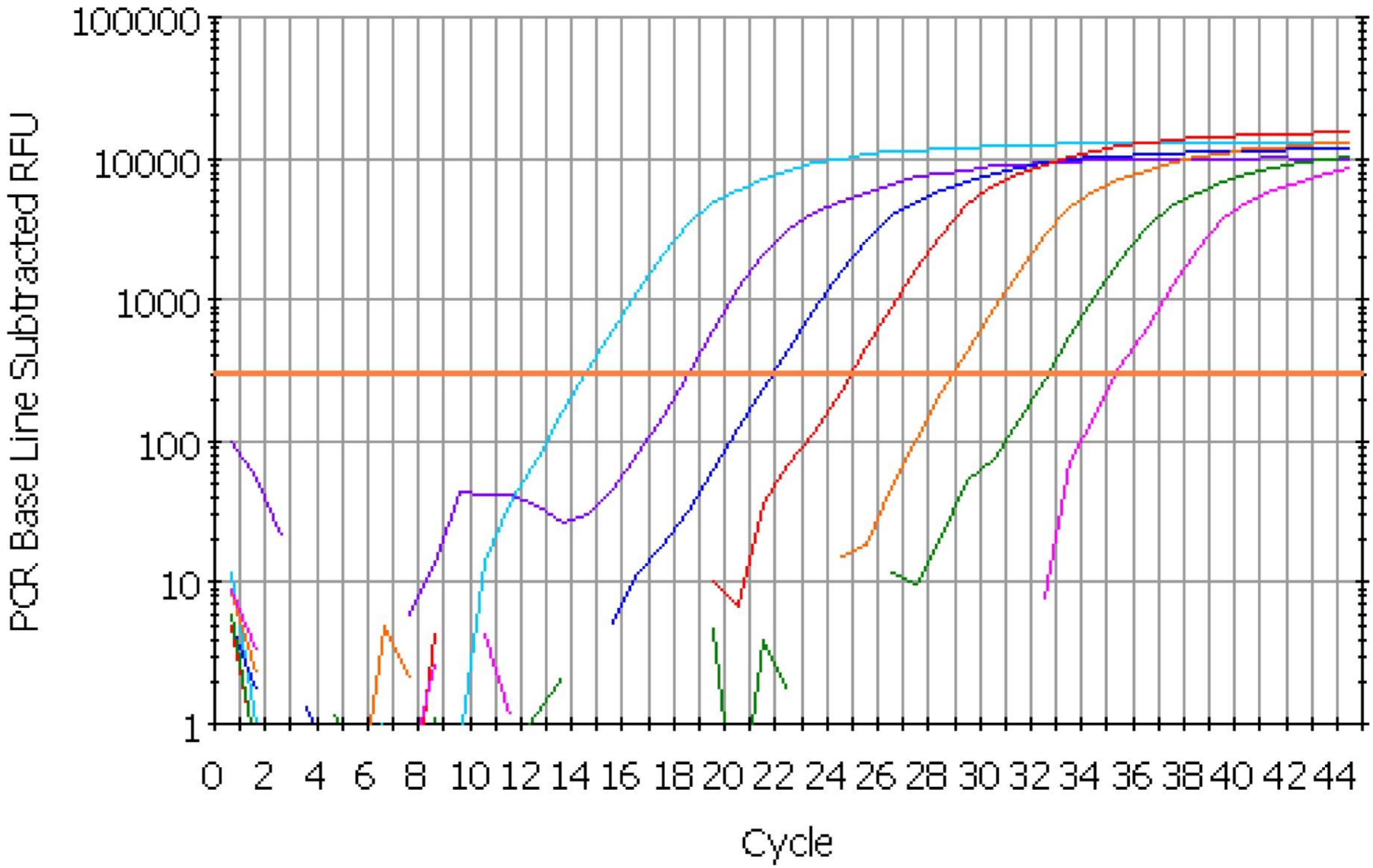
PCR efficiency

- In the ideal case with PCR we obtain a perfect duplication of the DNA at every thermal cycle: $qDNA = Ci * 2^N$
- In reality, the efficiency is not perfect due to various factors:
  - Not perfect reagents (inefficient enzymes, degraded bases etc.)
  - Not perfect thermal control (i.e. inhomogeneity)
  - Replication errors
  - Inhibitors thht interfere with the reaction
- In this case we define PCR efficiency:
  - 100% $qDNA = Ci * 2^N$
  - 80% $qDNA = Ci * 1.8^N$
  - 60% $qDNA = Ci * 1.6^N$
  - 0% $qDNA = Ci * 1^N$

Dilution series



…In log scale

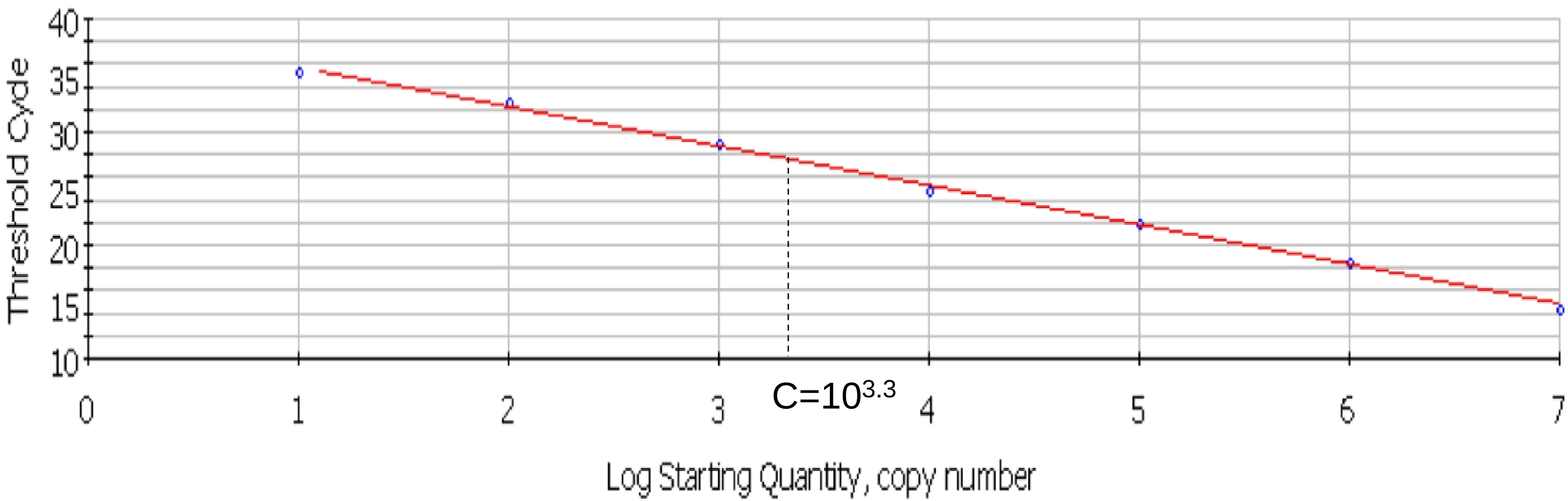

PCR Base Line Subtracted RFU
100000
10000
1000
100
10
1
0 2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44
Cycle

Cycle threshold

- An arbitrary threshold is defined for the amount of DNA.
- The cycle when the amount of DNA becomes greater then the threshold is named Cycle threshold (C_T)

- In practice, C_T can also be a non-integer number, obtained by interpolating the discrete PCR curve.

Realtime PCR linearity



- The value of CT varies linearly with the log of the initial concentration of DNA, on several orders of magnitude.
- Knowing the CT (and the calibration curve) it is possible to obtain the value of the initial concentration

Unknown sample → experiment: Ct=27.2

Variability of C_T

- The CT depends upon:
  - The value chosen for the threshold
  - The efficiency of the amplification
  - The accuracy of the measurement of the DNA concentration

- To obtain an absolute quantification, a few samples at known concentrations are amplified together with the unknown sample during the same experiment.

- The known samples are used to obtain the calibration curve, and the sample is quantified by comparison

Optical measurement of DNA

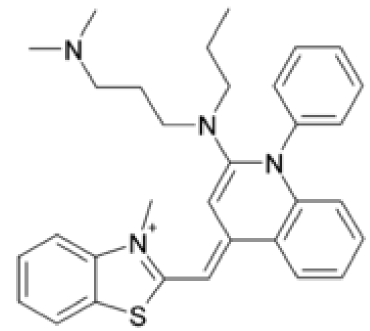
- The DNA concentration is linked to a flueorescence emission that is measured using standatd optical techniquies
- Several options:
  - Sybr green
  - TaqMan probes
  - Molecular beacons
  - Scorpion probes

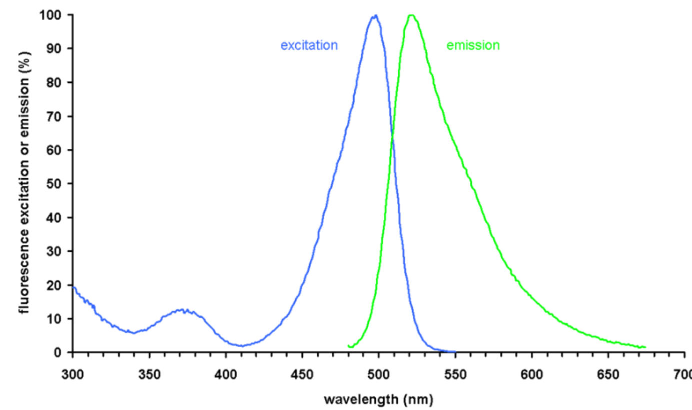
DNA Intercalants

- Are molecules that fits inside the grooves of the DNA double helic
- Normally they are not fluorescent. They emit fluorescence only when coupled to DNA
- In a solution with excess of intercalants, the measure of fluorescence is directly proportional to the DNA concentration.

Etidium bromide

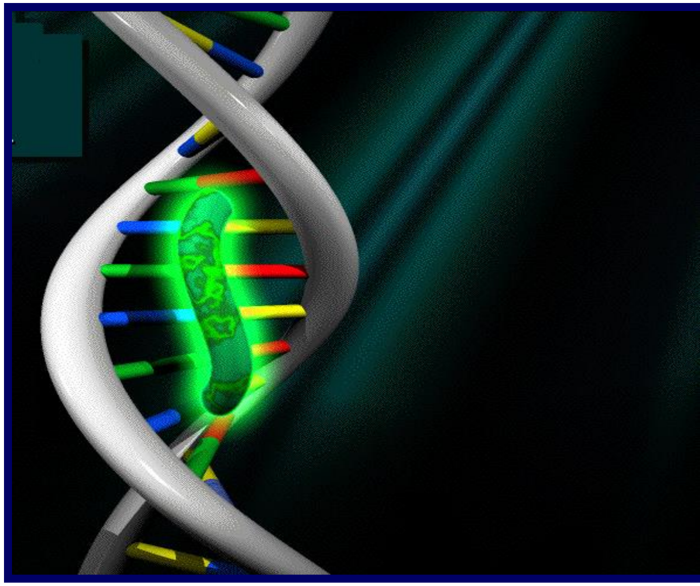
SYBR Green



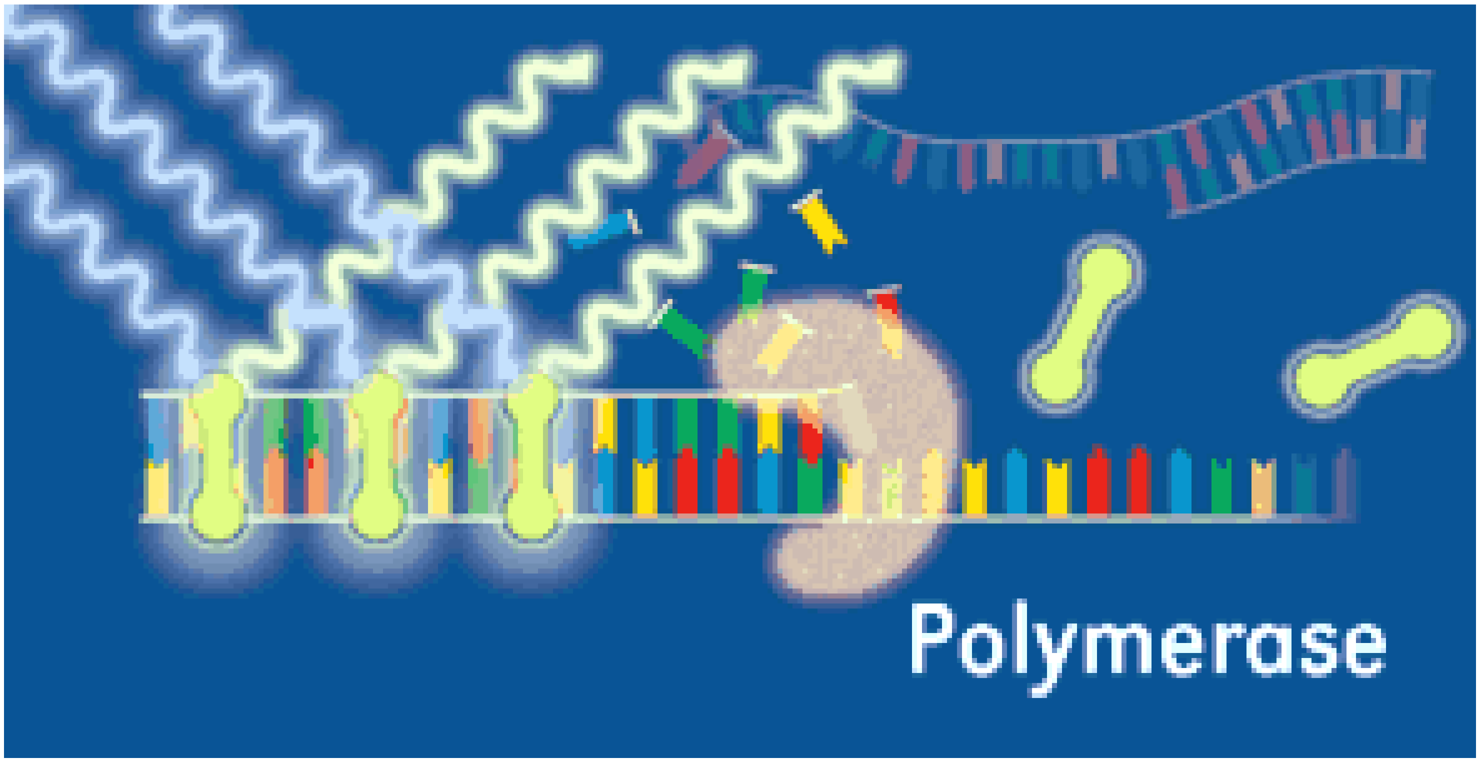


SYBR Green: principle

- Uses a non-specific fluorescent molecule that binds to the minor groove of DNA.
- SYBR green emits fluorescence only if a DNA double strand it is present .
- It detects only te total amount of DNA, it can’t distinquish between defferent sequences

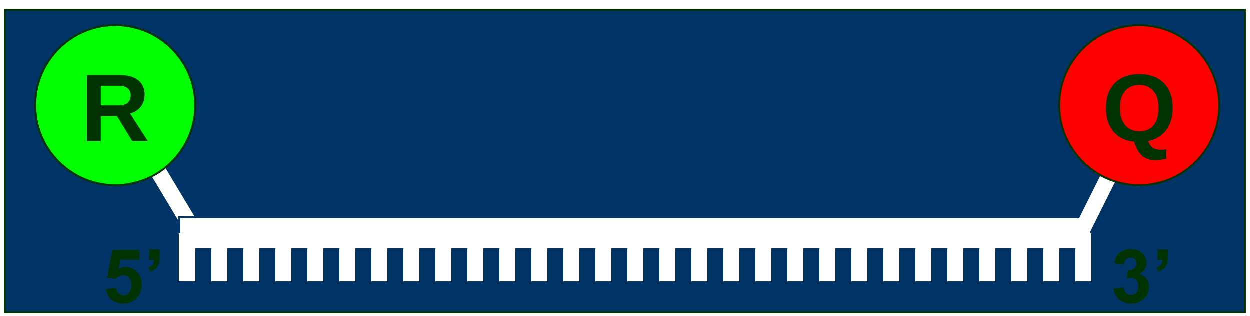


- During elongation phase an increase in fluorescence corresponds to the increase of the copies of the amplicone

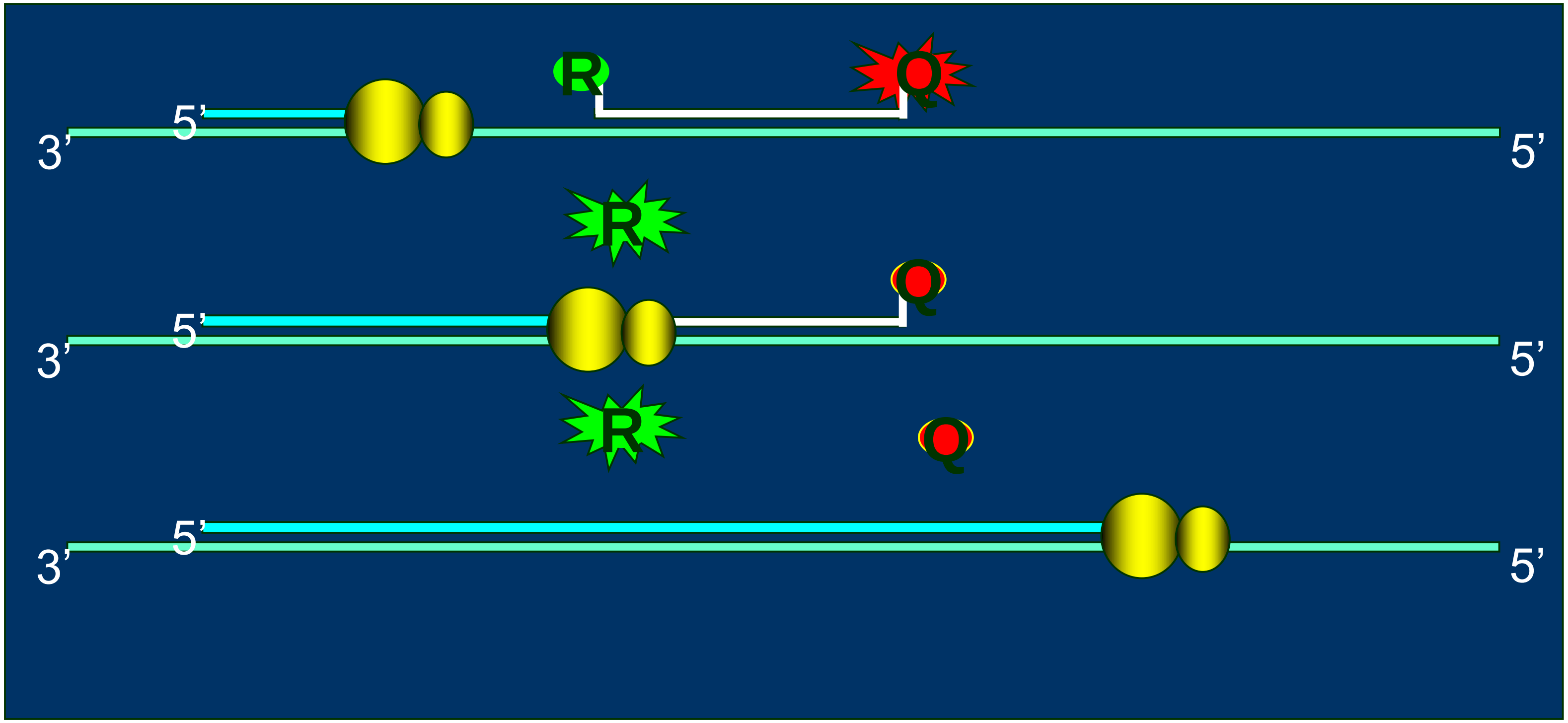


TaqMan probes

- Is a chain of normal DNA with a fluorophore linked at its 5’ end (Reporter) and another molecule (Quencer) in its 3’ termination
- The property of the quencher is to ‘switch off’ the fluorescence of R when Q and R are close each-other. The photons are absorbed by Q instead of being emitted by R.



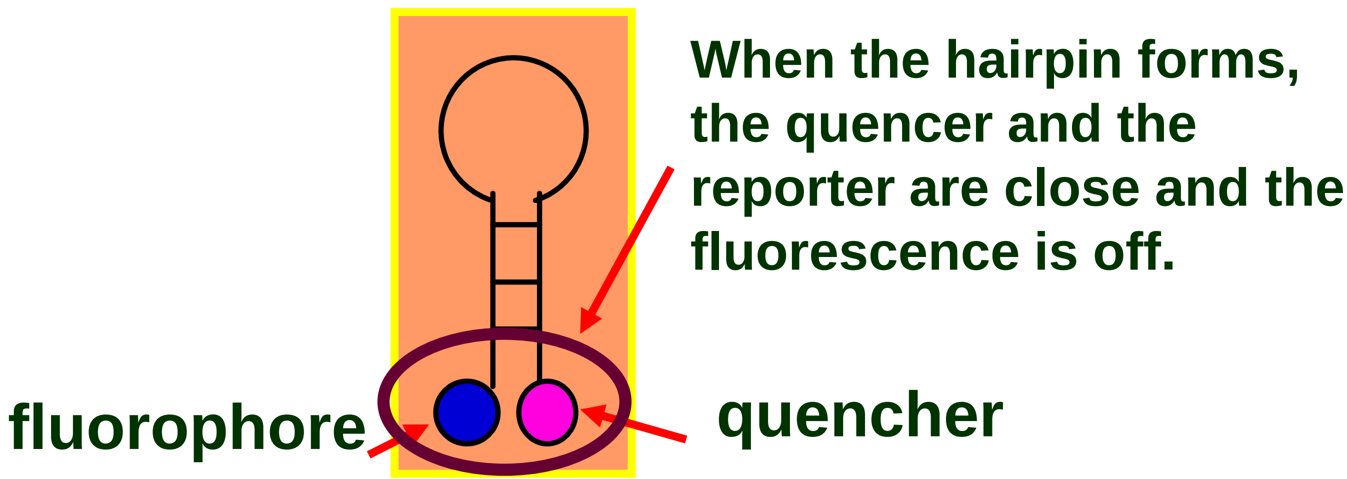
Real-Time PCR: 5'>3' eso-nucleasic activity



- The Polimerase enzyme during replication cuts out any existing DNA on its trail
- For any duplication event a new probe becomes fluorescent

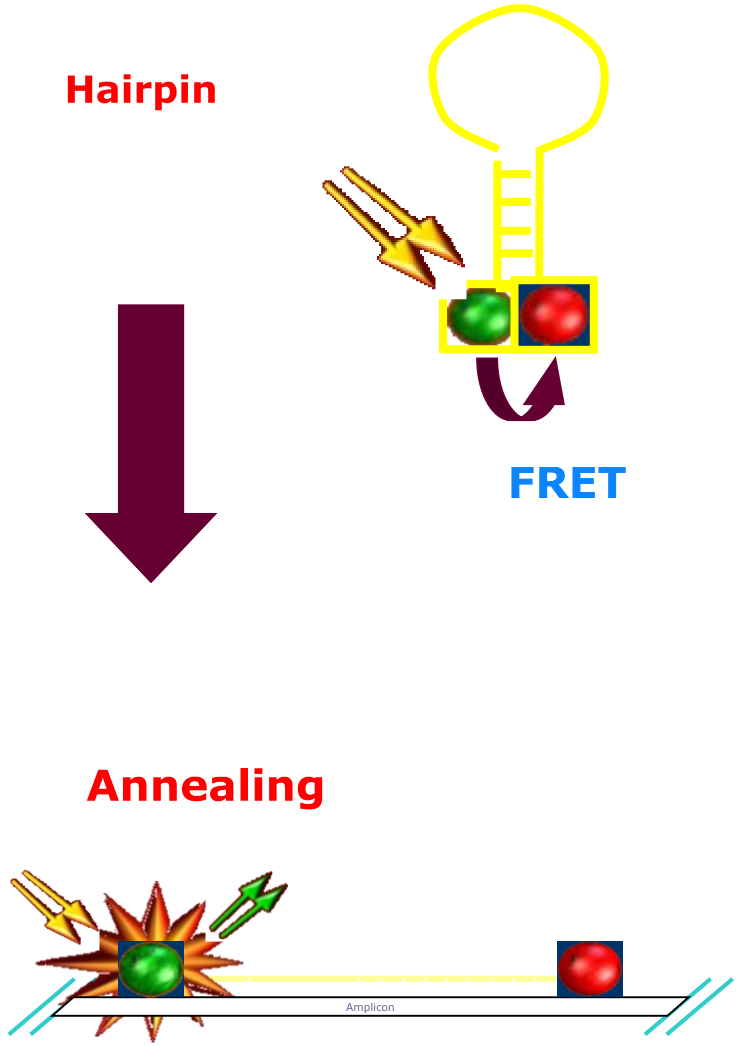
Molecular Beacons

- The "molecular beacons" have a fluorophore and a quencher at opposite ends of a small ssDNA (oligo nucleotide)
- The oligonucleotide is designed to be self-complementary, so the extremities form a double helix.
- The structure it forms is named hair-pin
- The central part of the DNA is complementary to the target



Molecular Beacons

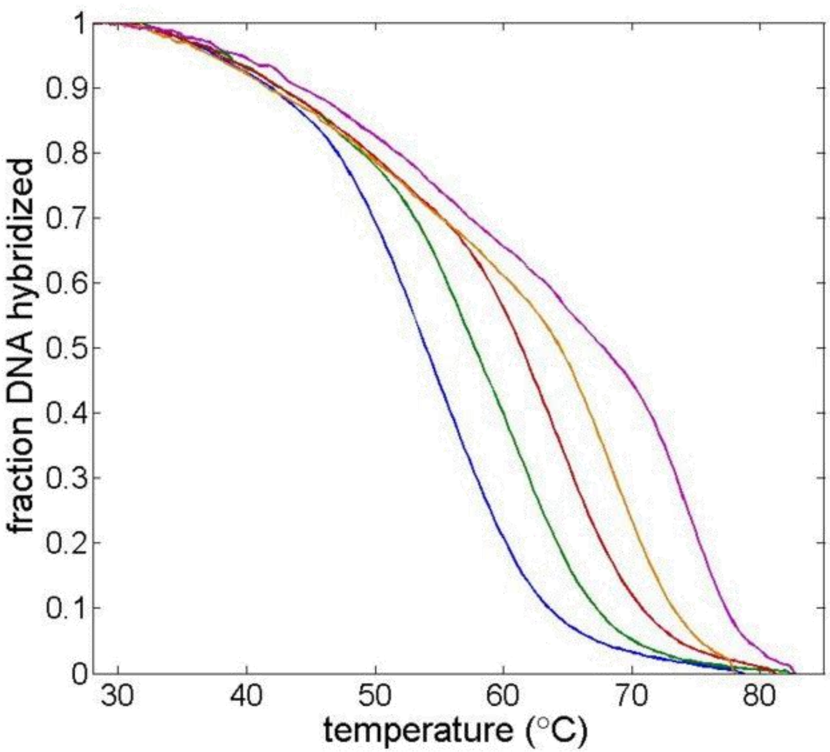
- After annealing step, the probe hibridize to the target, stretching linear. This spaces the Quencher apart producing a fluorescence signal.
- The amount of Total fluorescence depends on the quantity of specific DNA present in solution.
- Unlike Taqman, Beacon probes are not degraded during reaction, so can return to the hairpin state and switch-off

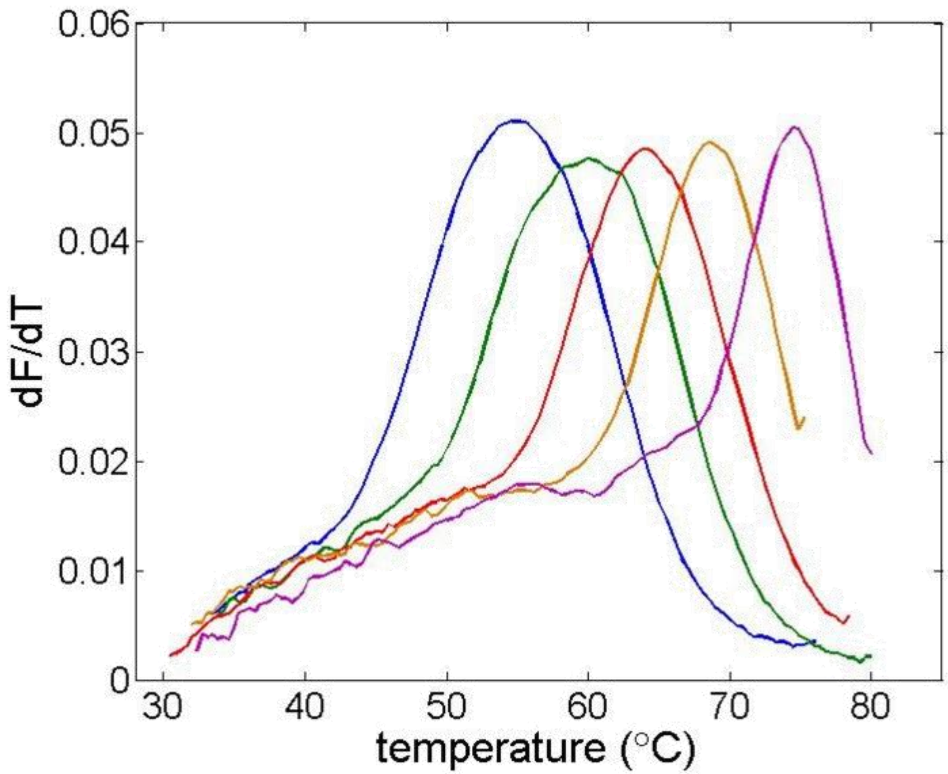


Melting curve

- After amplification, the system is slowly heated (thermal ramp) while the fluorescence is measured continuously.
- If a molecular beacon or an intercalant is present, when the DNA double helix denaturates, the fluorescence exhibits a negative peak.
- Dissociation temperature depends on
  - Amplicone length
  - CG percentage (3 H-bonds) over AT percentage (2 H-bonds)
- The (negated) derivative of the fluorescence is the most used indicator
- Melting curve can’t be done with TaqMar probes

Melting curves





Specificity of realtime PCR

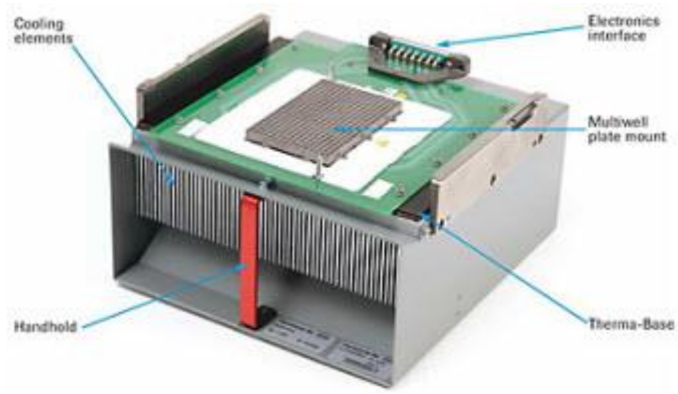
- With intercalants, the specificity is guaranteed only by the primers.
  - If in the sample is present DNA complementary to the primer couple, amplification happens and fluorescence is measured.
  - A melting curve after amplification gives clues about the DNA between primers (AT/GC, length)
- With TaqMan and molecular beacons it is possible to look for specific sequences inside the amplicone.
  - It is possible to mix several PCR in parallel on the same reactor.
  - Each reaction is measured by a different ‘color’, by selecting a different reporter for each taqMan probe.

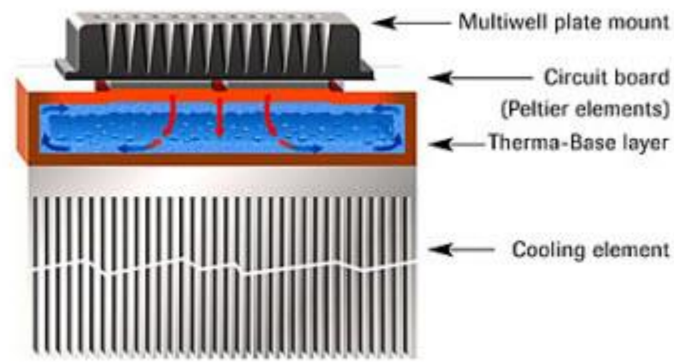
Quantification of RNA

- For gene expression assays it is important quantify the RNA content, related to cellular activity.
- RNA <u>can't </u>be directly amplified.
- An enzyme (reverse transcriptase) is used, before PCR. It translates RNA to DNA maintaining the same sequence.
- Once the DNA is obtained, it is amplified normally,

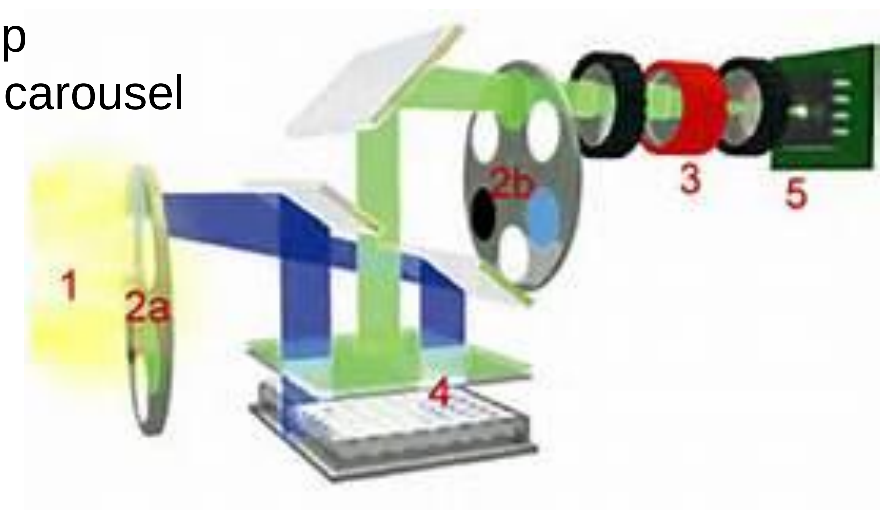
Typical Realtime PCR system components

Peltier cooled
allumininium plate

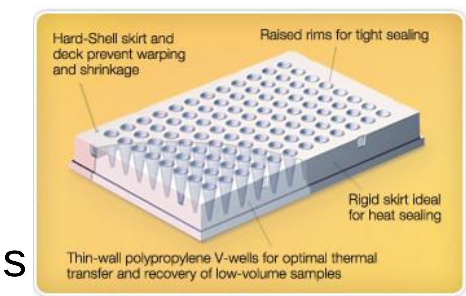




Halogen lamp
Moving filter carousel
CCD sensor



Polypropylene wells



Q3 system

Instrument

Disposable cartridge

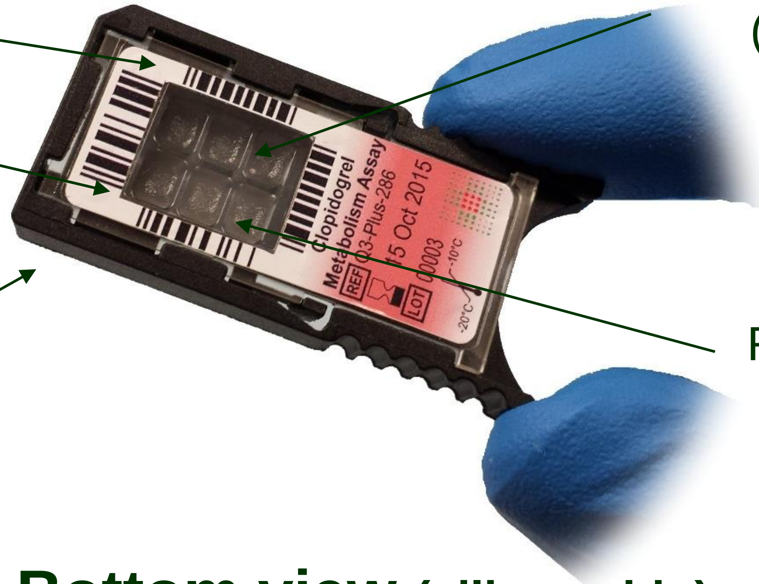
PCR on chip: the concept

Top view

Transparent plastic cover

Reagents under wax
(self-sealing)

Label with barcode

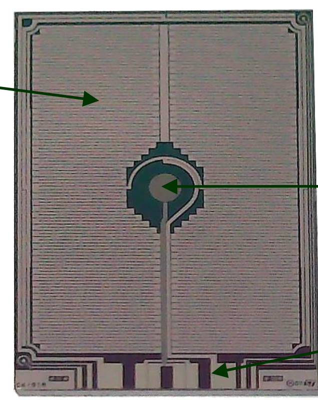


Silicon chip
+ RFID

Reaction chambers

Bottom view (silicon chip)

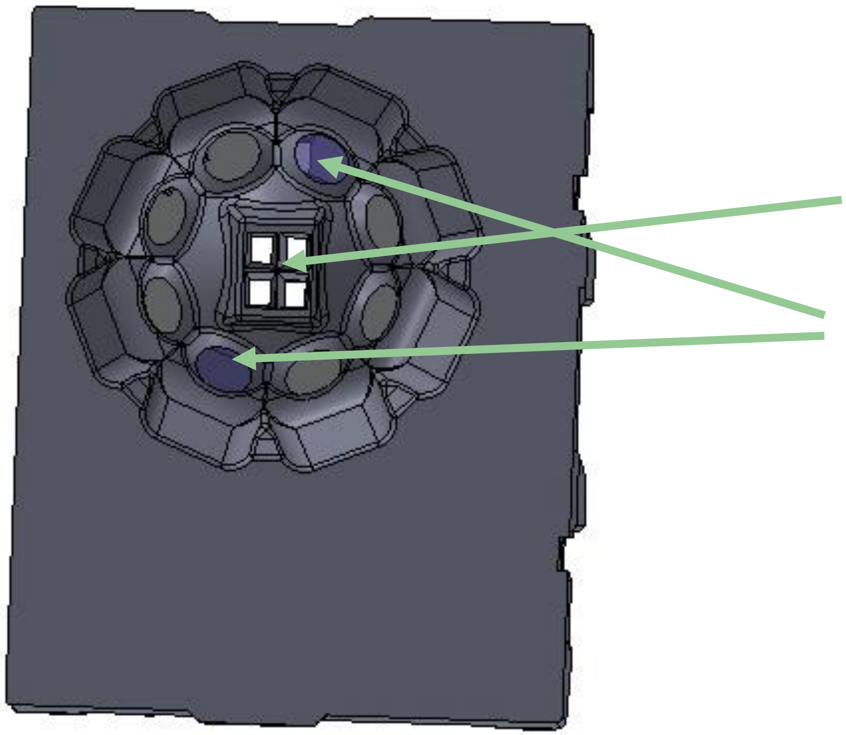
Heaters



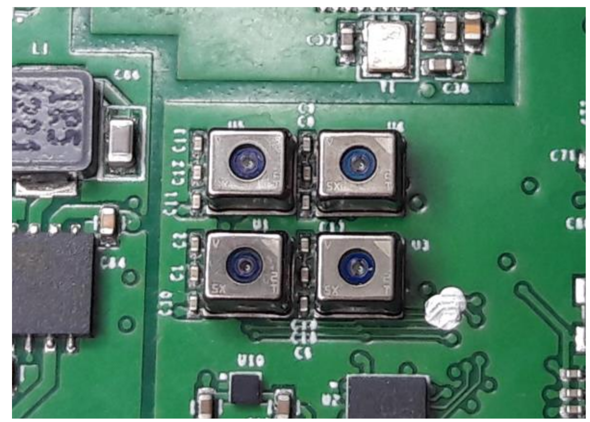
Temperature sensor

Direct contacts on chip pads

Optical head



- 4 CMOS image sensors, closely placed, looking at the same area.
- 4 pairs of LED illuminating the chip



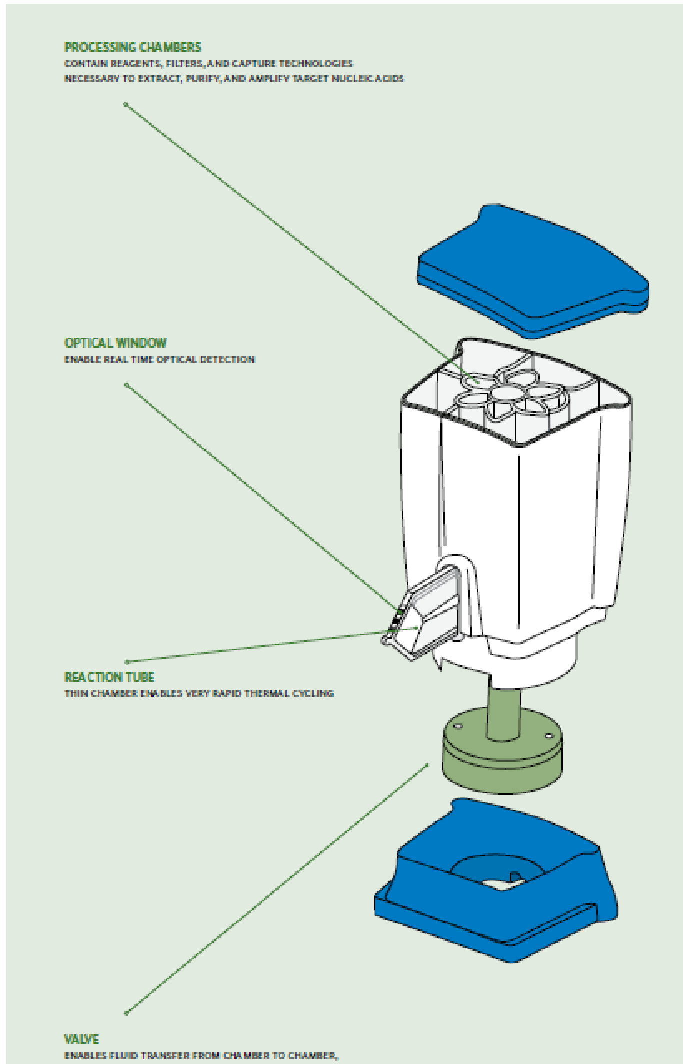
- Each sensor and pair of LED is behind a different optical filter
- Instead of moving a filter carousel, different LED/Sensors are switched on

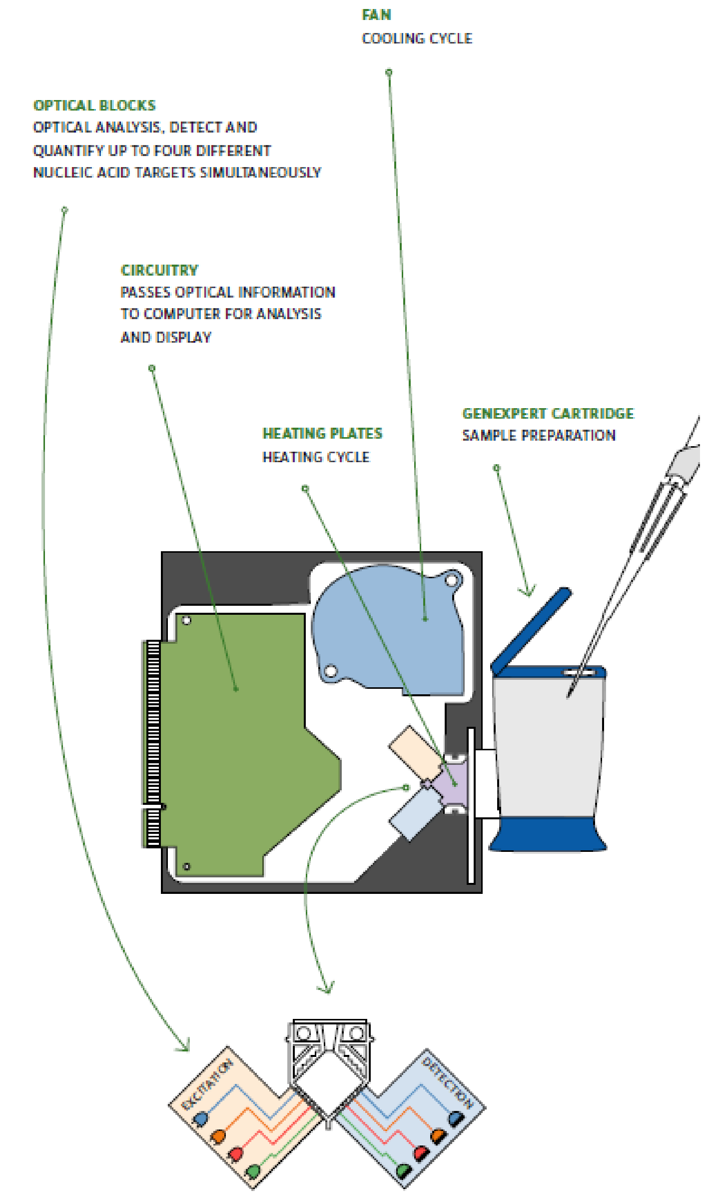
Genexpert (Cepheyd)

- The first commercial system that integrates Sample Preparation and Realtime PCR
- Disposable cartridges ready to use, filled with reagents specific for each application.
- User only need to insert the sample (es. Blood) and start the system.
- Sample-to-answer in about an hour

Principle

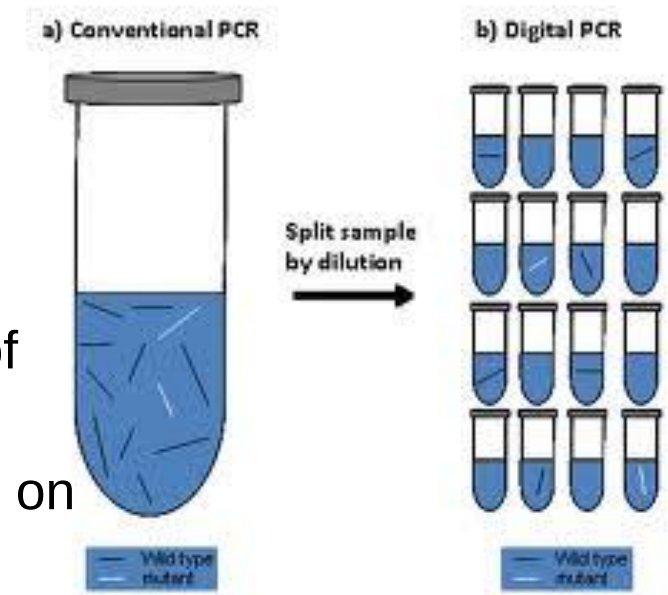
- Liquid handling by valves and pistons, activated by the instrument.
- Heating and cooling by air flow.
- Optical detection at 4 channels





Digital PCR

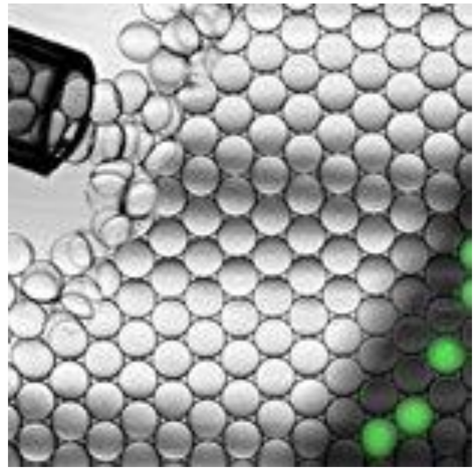
- Extremely precise: it can quantify the Exact number of copies of DNA

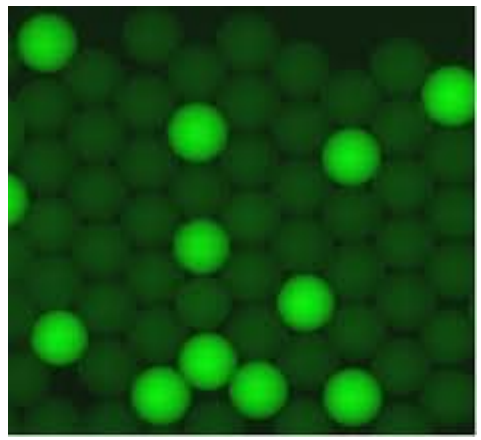


- The initial sample + PCR mix is fractioned into a huge number of subsamples. The number of subsamples is higher than the maximum number of expected copies.
- A standard (endpoint) PCR is performed in parallel on each subsample.
- If the target gene is present in the sample, it becomes fluorescent, otherwise it remains dark.
- The number of subsamples on is exactly the number of copies present initially

Digital PCR in emulsion

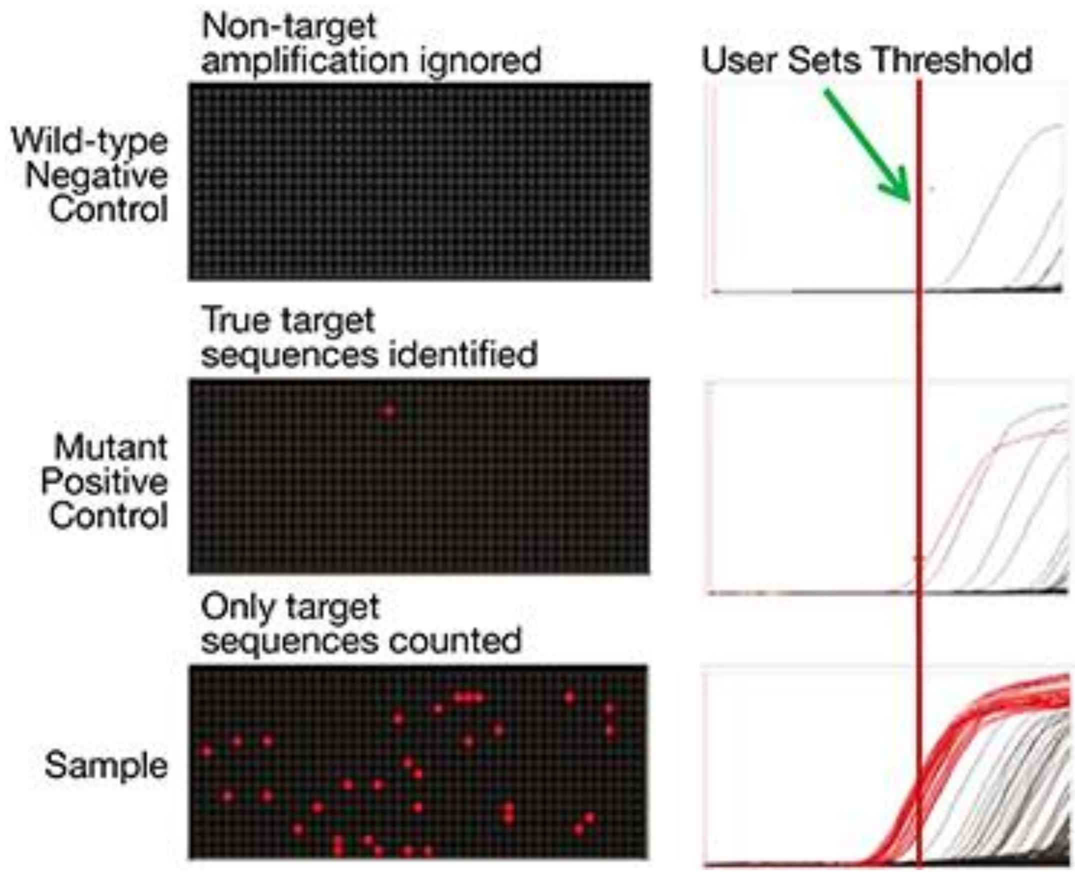
- ‘droplets’ generation, small drops of mix in mineral oil http://www.youtube.com/watch?v=1Pj9pH24NF0
- The whole emulsion undergoes a PCR reaction in a single vial. The emulsion must be stable in order for the droplets not to mix.
- After the PCR the ON droplets are counted by an automatic counter





Digital PCR on chip

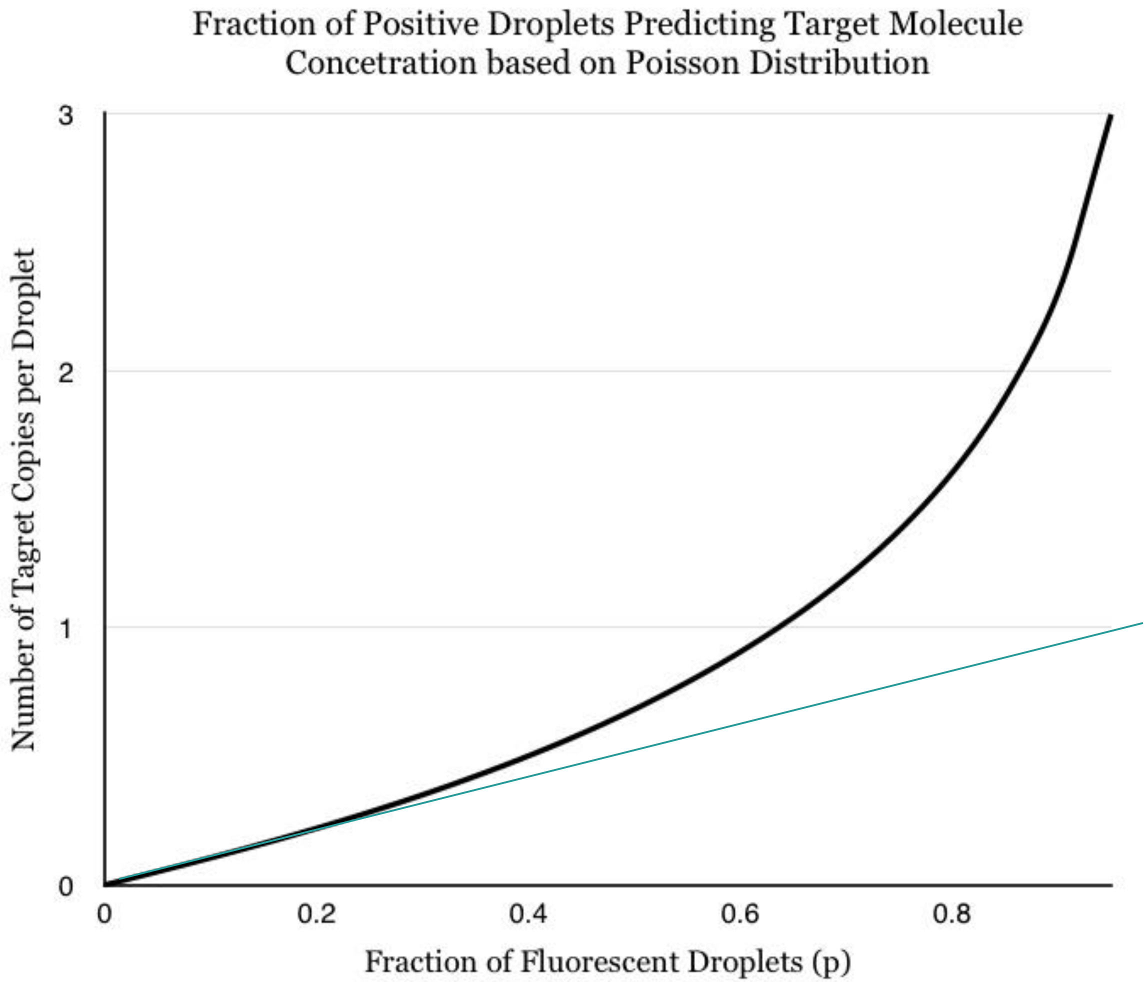
- Fluidigm



Detection limits (dynamic range)

- Nc=number of DNA copies, Nd=number of droplets
- The probability of occurrence of droplets with 0,1,2,… copies follows a poissonian distribution.
- When Nd>>Nc, the probability of having 2 copies in the same droplets is negligible, so the ON droplets number equals the number of copies.
- When Nd~Nc, the probability of having 2 or more copies of DNA in the same droplet is no more negligible, so an appropriate correction factor must be applied.

Percentage of positives vs. concentration



Isothermal amplification

- After PCR, several methods have been invented that doesn't require temperature change to amplify DNA
- Among them, LAMP (Loop-mediated isothermal amplification) is the most known

- Advantages:
  - Very simple instrumentation, just a stable heater, typ 60C
  - Faster than PCR, typ 10~15 min (continuous time)
- Disadvantages
  - Not quantitative (no cycles → no Ct → no calibration curve → no concentration)

Ok for rapid tests